

# The maximal energy of integral circulant graphs of orders $p^{2r}3^{2s+1}$ , $pq^m$ and $p^2q^{b*}$

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## Abstract

Jiang and Yang determined the maximal energy of integral circulant graphs of order  $n = p^{2r}q^{2s+1}$ , where  $p, q$  are distinct odd primes with  $q \geq 5$ : the checkerboard divisor set  $\{p^i q^j : i+j \text{ even}\}$  is the unique maximiser. They noted that their semidefinite method does not extend to  $q = 3$ . We prove the remaining case: for every prime  $p \geq 5$  and all  $r \geq 1, s \geq 0$ , the checkerboard set is the unique energy maximiser for order  $p^{2r}3^{2s+1}$ , with the maximal energy given by the formula of Jiang and Yang. This settles the maximal-energy problem for opposite-parity exponents for all pairs of distinct odd primes. The key step is a sharp inequality for sign matrices under weighted Ramanujan transforms. We prove it with a path identity and an explicit potential function, and the same argument gives a semidefinite-free proof of the theorem of Jiang and Yang. For exponents of equal parity the checkerboard set contains  $n$  and is not admissible. We conjecture the maximal energy in this case, with two maximisers, and prove the conjecture when one of the exponents is 1 or 2 and  $p, q$  are any distinct odd primes; for  $n = pq$  it also follows from known energy formulas. For example, for  $n = p^2 q^b$  with  $b$  even the maximal energy is  $\frac{1}{2}[n + (5p^2 - 8p + 4)d_b(q)] - (p^2 - 2p + 2)\delta_b(q)$ , where  $d_b$  and  $\delta_b$  are explicit polynomials, and it is attained exactly by  $\{p^i q^j : i+j \text{ odd}\}$  and  $\{p^i q^j : i+j \text{ even}\} \setminus \{n\}$ . With the opposite-parity case, this determines the maximal energy for every order  $p^a q^b$  with  $\min\{a, b\} \leq 2$ . A computer-assisted extension of the argument, whose certificates were checked by three independently written programs, proves the conjecture also when one of the exponents is at most 8. The main theorem, the two sign-matrix inequalities, the energy formula for these graphs and the theorems for exponents 1 and 2 are formally verified in Lean 4; the closed-form evaluation of the maximal energies and the computer-assisted results are not formalised. The proofs were found by AI coding agents working under the author's direction; their role is described at the end of the paper.

## 1 Introduction

The energy of a finite graph is the sum of the absolute values of its adjacency eigenvalues [2]. For  $n \geq 2$  and a nonempty set  $\mathcal{D}$  of proper divisors of  $n$ , the integral circulant graph  $\text{ICG}_n(\mathcal{D})$  has vertex set  $\mathbb{Z}/n\mathbb{Z}$ , two distinct vertices  $u, v$  being adjacent when  $\gcd(u - v, n) \in \mathcal{D}$ . These are the circulant graphs with integral spectrum [11]. The problem of maximising the energy of  $\text{ICG}_n(\mathcal{D})$  over  $\mathcal{D}$  for fixed  $n$  has been solved for prime powers [9] and studied for other orders; see the introduction of [6] for a survey. Roldán [8] conjectured that for  $n = p^2 q^3$  the checkerboard set  $\{1, p^2, pq, q^2, p^2 q^2, pq^3\}$  is the unique maximiser. Jiang and Yang [6] proved much more.

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\*Version 1.5 extends Theorem 5 to  $\min\{a, b\} \leq 8$ ; the other results and the numbering of version 1.4 are unchanged. Version 1.4, released as version 1.4.0 of [1], added Theorems 4 and 5 (exponents of equal parity, one of them at most 4) and Sections 10–12 to version 1.2 (release 1.2.0). It renumbered the statements after Theorem 3 and moved the section on verification to Section 13; Theorem 10, Conjecture 12 and Proposition 13 of version 1.2 are now Theorem 12, Conjecture 14 and Proposition 15. The results of version 1.2 are unchanged.

**Theorem 1** (Jiang–Yang [6, Theorem A]). *Let  $p, q$  be distinct odd primes with  $q \geq 5$ , let  $r \geq 1$ ,  $s \geq 0$  and  $n = p^{2r}q^{2s+1}$ . Among all nonempty sets of proper divisors of  $n$ , the energy of  $\text{ICG}_n(\mathcal{D})$  is uniquely maximised by*

$$\mathcal{D}_{r,s}^* = \{p^i q^j : 0 \leq i \leq 2r, 0 \leq j \leq 2s+1, i+j \text{ even}\},$$

*and the maximal energy is  $\frac{1}{2}[n + d_{2r}(p) d_{2s+1}(q)]$ , where  $d_k(x) = (2k+1)x^k + 4 \sum_{j=0}^{k-1} (-1)^{k-j}(j+1)x^j$ .*

They remark that the case  $q = 3$  is not covered, because their odd-exponent semidefinite argument fails at  $x = 3$  [6, Remark 4.2]. Our main result removes this restriction.

**Theorem 2.** *The conclusion of Theorem 1 holds for  $q = 3$ : for every prime  $p \geq 5$  and all  $r \geq 1$ ,  $s \geq 0$ , the set  $\mathcal{D}_{r,s}^*$  is the unique energy maximiser among nonempty sets of proper divisors of  $n = p^{2r}3^{2s+1}$ , and the maximal energy is  $\frac{1}{2}[n + d_{2r}(p) d_{2s+1}(3)]$ .*

Together with Theorem 1 (which allows  $p = 3$  on the even side), Theorem 2 determines the maximal energy and the unique maximiser for every order  $p^{\text{even}}q^{\text{odd}}$  with  $p, q$  distinct odd primes. The case  $r = s = 1$  was treated earlier. Park [7] claims a proof of Roldán’s conjecture for all distinct odd primes. Schreib [10] gives a written proof of the case  $q = 3$  and combines it with [6] to obtain the conjecture. An arithmetic certificate for  $n = 27p^2$  was announced in [12]. These works concern only the exponents  $(2, 3)$ , and we are not aware of any earlier treatment of the case  $q = 3$  for general  $r$  and  $s$ .

If the two exponents have the same parity, the checkerboard set contains  $n$  itself and is not a set of proper divisors, and the maximiser changes. For this case we state a conjecture (Conjecture 14), with two maximisers and an explicit maximal energy. We prove it when one of the exponents equals 1 or 2, and, with computer assistance, when one of them is between 3 and 8. For  $n = p^a q^b$ , a set written as  $\{p^i q^j : \dots\}$  consists of the divisors  $p^i q^j$  of  $n$  ( $0 \leq i \leq a$ ,  $0 \leq j \leq b$ ) with the stated property, and  $\delta_k(x) = (x^k(x-1) + 2(-1)^k)/(x+1)$  (Remark 9).

**Theorem 3.** *Let  $p, q$  be distinct odd primes, let  $m \geq 1$  be odd, and let  $n = pq^m$ . The maximal energy of an integral circulant graph of order  $n$  is*

$$\frac{1}{2}[n + (3p-4)d_m(q)] - (p-2)\delta_m(q),$$

*and it is attained exactly by the sets  $\{p^i q^j : i+j \text{ odd}\}$  and  $\{p^i q^j : i+j \text{ even}\} \setminus \{n\}$ . Exchanging the names of the primes, for  $n = p^m q$  the maximal energy is  $\frac{1}{2}[n + (3q-4)d_m(p)] - (q-2)\delta_m(p)$ , attained exactly by the two sets with the same description.*

For  $m = 1$  the maximal energy is  $4(p-1)(q-1)$ , attained by  $\{p, q\}$  and  $\{1\}$ . This case can also be derived from published energy formulas of Ilić [3] and of Ilić and Bašić [4] (Remark 16), although we have not found it stated. For the exponent 2 we prove the following.

**Theorem 4.** *Let  $p, q$  be distinct odd primes, let  $b \geq 2$  be even, and let  $n = p^2 q^b$ . The maximal energy of an integral circulant graph of order  $n$  is*

$$\frac{1}{2}[n + (5p^2 - 8p + 4)d_b(q)] - (p^2 - 2p + 2)\delta_b(q),$$

*and it is attained exactly by the sets  $\{p^i q^j : i+j \text{ odd}\}$  and  $\{p^i q^j : i+j \text{ even}\} \setminus \{n\}$ . Exchanging the names of the primes, for  $n = p^b q^2$  the maximal energy is  $\frac{1}{2}[n + (5q^2 - 8q + 4)d_b(p)] - (q^2 - 2q + 2)\delta_b(p)$ , attained exactly by the two sets with the same description.*

Here  $5x^2 - 8x + 4 = d_2(x)$  and  $x^2 - 2x + 2 = \delta_2(x)$ , so Theorems 3 and 4 are the cases  $\min\{a, b\} = 1$  and  $\min\{a, b\} = 2$  of Conjecture 14. Theorem 4 contains the exponent pairs  $(2, 2)$ ,  $(2, 4)$ ,  $(2, 6)$ ,  $(2, 8)$  and their transposes, which were previously covered only by the computer-assisted certificates of Proposition 15. Its proof gives a sign-matrix inequality for all real  $p \geq 5$ ,  $q \geq 3$  and for  $p = 3$ ,  $q \geq 5$  (Theorem 25). The same method, with a computer-assisted step, gives the following.

**Theorem 5.** *Let  $p, q$  be distinct odd primes, let  $a, b \geq 1$  with  $a + b$  even and  $\min\{a, b\} \leq 8$ , and let  $n = p^a q^b$ . The maximal energy of an integral circulant graph of order  $n$  is  $\frac{1}{2}[n + d_a(p) d_b(q)] - \delta_a(p) \delta_b(q)$ , and it is attained exactly by the sets  $\{p^i q^j : i + j \text{ odd}\}$  and  $\{p^i q^j : i + j \text{ even}\} \setminus \{n\}$ . For  $3 \leq \min\{a, b\} \leq 8$  the proof is computer-assisted, and it is not formalised in Lean.*

For  $\min\{a, b\} \leq 2$  this is Theorem 3 or Theorem 4. For  $3 \leq \min\{a, b\} \leq 8$ , the method of Theorem 4 reduces the claim, separately for each of these exponents and for all values of the other exponent at once, to a finite list of inequalities between explicit rational functions of  $p$  and  $q$ . We verified these lists by exact computer calculation with three independently written programs, two of them written by referee agents (Section 12). Proposition 15 gives an independent check of eleven of the exponent pairs.

Theorems 1 and 2 cover every order whose exponents have opposite parity. So Theorems 1–5 determine the maximal energy, and all maximisers, for every order  $p^a q^b$  with  $\min\{a, b\} \leq 8$ , where  $p, q$  are distinct odd primes; for  $\min\{a, b\} \leq 2$  the proof does not depend on a computer. The conjecture remains open when  $\min\{a, b\} \geq 9$ . Our searches for earlier work on exponents of equal parity were made on 26 September 2026, between 03:37 and 04:32 UTC, in arXiv, OpenAlex, Semantic Scholar, Google Scholar, web search engines, and GitHub code, issues, pull requests and commits, including the repositories of [7, 10, 12]; an incremental search of arXiv, OpenAlex and GitHub followed between 05:53 and 05:58 UTC. They found no earlier determination of the maximal energy for such orders; for  $n = pq$  it can be derived from known energy formulas (Remark 16). A final check on 26 September 2026 between 15:49 and 15:52 UTC, repeated for version 1.5 between 19:04 and 19:06 UTC (the arXiv, including the versions of [6]; GitHub, including the issues of [12]; MathDB; and MathOverflow, through the Stack Exchange API) found nothing new. This reports the coverage of our searches, not a guarantee of priority.

Our proof replaces the semidefinite part of [6] by Theorem 12 below, an inequality for sign matrices that holds for all exponents and parities and all real  $p \geq 5, q \geq 3$ . By symmetry it also yields Theorem 1 without semidefinite certificates (Remark 13). Section 8 treats exponents of equal parity, and Section 9 proves Theorem 3 by applying the path identity and the potential of Sections 3 and 4 to the two rows of the sign matrix. Section 10 applies them to all rows of  $T_a(p)Y$  at once, so that the states of the path are vectors built from the columns of the sign matrix  $Y$ . With this, Section 11 proves Theorem 4 and Section 12 proves Theorem 5.

## 2 Weighted Ramanujan transforms

We use the notation of [6]. For real  $x > 1$ , an integer  $k \geq 1$  and  $0 \leq i, j \leq k$ , let  $T_k(x) = (t_{ij})$  be the symmetric matrix with

$$t_{ij} = \begin{cases} 0, & i + j \leq k - 2, \\ -x^{k-1}(x - 1), & i + j = k - 1, \\ x^{2k-i-j-2}(x - 1)^2, & i + j \geq k, \end{cases} \quad (i, j < k),$$

$t_{ik} = t_{ki} = x^{k-i-1}(x - 1)$  for  $i < k$ , and  $t_{kk} = 1$ . When  $x$  is prime,  $t_{ij} = \varphi(x^{k-i}) c_{x^{k-j}}(x^i)$ , where  $c_m(u)$  denotes Ramanujan's sum [6, (2.2)–(2.4)]. The row sums are  $T_k(x)\mathbf{1} = x^k e_k$  [6, (2.5)], where  $e_k$  is the last standard basis vector. For a real matrix  $Z$  let  $\|Z\|_1$  be the sum of the absolute values of its entries.

**Proposition 6** ([6, Proposition 2.2]). *Let  $p \neq q$  be primes,  $a, b \geq 1$ ,  $n = p^a q^b$ , and let  $X \in \{0, 1\}^{(a+1) \times (b+1)}$  have  $x_{ij} = 1$  exactly when  $p^i q^j \in \mathcal{D}$ . Then  $\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \|T_a(p)XT_b(q)^\top\|_1$ .*

Let  $s_k = (1, -1, 1, \dots, (-1)^k)^\top$  and define  $d_k(x) = \|T_k(x)s_k\|_1$ , and put  $d_0(x) = 1$ . By Remark 9 below,  $d_k(x)$  agrees with the closed form in Theorem 1 for all real  $x > 2$ ; see also [6, (3.6)].

### 3 A path identity

For  $w \in \mathbb{R}^{k+1}$  define  $z_{k-1} = w_k$  and, for  $j = k-1, k-2, \dots, 0$ ,

$$z_{j-1} = \frac{z_j + (x-1)w_j}{x}. \quad (1)$$

Thus  $z_j$  is a convex combination of  $w_{j+1}, \dots, w_k$ , and each  $z_j$  is a linear function of  $w$ .

**Lemma 7.** For  $w \in \mathbb{R}^{k+1}$  we have

$$\|T_k(x)w\|_1 = x^k \left( |z_{-1}| + \sum_{j=0}^{k-1} |z_{j-1} - z_j| \right).$$

*Proof.* Unwinding (1) gives  $z_j = x^{-(k-j-1)} \sum_{i>j} \varphi(x^{k-i})w_i$  with  $\varphi(x^0) = 1$  and  $\varphi(x^e) = x^{e-1}(x-1)$ . The last row of  $T_k(x)$  is  $(\varphi(x^{k-i}))_i$ , so  $(T_k(x)w)_k = x^k z_{-1}$ . For  $u < k$  put  $j = k-1-u$ . Row  $u$  has entry  $-x^{k-1}(x-1)$  in column  $j$ , entries  $x^j(x-1)\varphi(x^{k-i})$  in columns  $i > j$ , and zeros elsewhere. Hence  $(T_k(x)w)_u = x^{k-1}(x-1)(z_j - w_j)$ , while  $z_{j-1} - z_j = \frac{x-1}{x}(w_j - z_j)$ .  $\square$

### 4 The one-dimensional potential

For  $x > 2$  let  $\mu_{-1} = 1$ ,  $\mu_j = (x-1-\mu_{j-1})/x$  for  $j \geq 0$ , and  $c_j = (x-1)(1+\mu_{j-1})/x$ . By induction,  $\mu_j \in [\frac{x-2}{x}, 1]$  for all  $j$ .

**Lemma 8.** Let  $x > 2$ , let  $y \in \{\pm 1\}^{k+1}$ , and let  $z_j$  be defined by (1) with  $w = y$ . Then  $\text{sgn } z_j = y_{j+1}$  and  $\frac{x-2}{x} \leq |z_j| \leq 1$  for  $-1 \leq j \leq k-1$ , and

$$\frac{\|T_k(x)y\|_1}{x^k} = \sum_{j=0}^{k-1} c_j + \mu_{k-1} - \sum_{j \in \text{NA}(y)} 2\mu_j |z_j|, \quad \text{NA}(y) = \{0 \leq j \leq k-1 : y_j = y_{j+1}\}.$$

Consequently  $\|T_k(x)y\|_1 \leq d_k(x)$ , with equality exactly when  $y = \pm s_k$ , and  $d_k(x) = x^k(\sum_j c_j + \mu_{k-1})$ . Moreover  $(T_k(x)s_k)_k > 0$ .

*Proof.* We have  $z_{k-1} = y_k$ , and since  $x-1 > 1 \geq |z_j|$ , the sign of  $z_{j-1} = (z_j + (x-1)y_j)/x$  is  $y_j$  and  $|z_{j-1}| = (x-1 \pm |z_j|)/x \in [\frac{x-2}{x}, 1]$ . Put  $f_j = |z_{j-1} - z_j| + \mu_{j-1}|z_{j-1}| - \mu_j|z_j|$ . If  $y_j = -\text{sgn } z_j$ , then  $|z_{j-1} - z_j| = \frac{x-1}{x}(1+|z_j|)$  and  $|z_{j-1}| = \frac{x-1-|z_j|}{x}$ ; using  $x\mu_j = x-1-\mu_{j-1}$  one finds  $f_j = c_j$ . If  $y_j = \text{sgn } z_j$ , the same computation with  $|z_{j-1} - z_j| = \frac{x-1}{x}(1-|z_j|)$  and  $|z_{j-1}| = \frac{x-1+|z_j|}{x}$  gives  $f_j = c_j - 2\mu_j|z_j|$ . Summing, the terms  $\mu_j|z_j|$  telescope, and Lemma 7 gives the identity, since  $\mu_{-1} = 1$  and  $|z_{k-1}| = 1$ . Every  $j \in \text{NA}(y)$  contributes a positive term  $2\mu_j|z_j|$ , and  $\text{NA}(y) = \emptyset$  exactly for  $y = \pm s_k$ . Finally  $(T_k(x)s_k)_k = x^k z_{-1}$  and  $\text{sgn } z_{-1} = (s_k)_0 = 1$ .  $\square$

*Remark 9.* Put  $\delta_k(x) = x^k \mu_{k-1}$ . Solving the recursion for  $\mu$  gives  $\delta_k(x) = (x^k(x-1) + 2(-1)^k)/(x+1)$ . For  $y = s_k$ , every step in the proof of Lemma 8 has  $y_j = -\text{sgn } z_j$ , so  $|z_{j-1}| = (x-1-|z_j|)/x$ ; starting from  $|z_{k-1}| = 1 = \mu_{-1}$  this gives  $|z_j| = \mu_{k-2-j}$ , and in particular  $(T_k(x)s_k)_k = x^k z_{-1} = \delta_k(x)$ . Since the potential does not depend on  $k$ , Lemma 8 gives  $d_{k+1}/x^{k+1} - d_k/x^k = c_k + \mu_k - \mu_{k-1} = 2\mu_k$ , that is,  $d_{k+1}(x) = x d_k(x) + 2\delta_{k+1}(x)$ , with  $d_0 = 1$ . The closed form  $(2k+1)x^k + 4 \sum_{j=0}^{k-1} (-1)^{k-j}(j+1)x^j$  satisfies the same recursion and initial value, so it equals  $d_k(x)$  for all  $k \geq 0$ .

## 5 A two-variable inequality

**Lemma 10.** *Let  $P \geq \mu$  and  $0 \leq \mu \leq 1$ , and put  $h(A, B) = P|A - B| + \mu|B + PA| - (P - \mu)|B| - P(1 + \mu)|A|$  for real  $A, B$ . Then  $h(A, B) \leq 2\mu \max\{0, |B| - P|A|\}$ .*

*Proof.* As  $h(-A, -B) = h(A, B)$  we may assume  $A \geq 0$ . If  $0 \leq B \leq A$  then  $h = -2(P - \mu)B \leq 0$ . If  $B > A$  then  $h = 2(\mu B - PA) \leq 2\mu(B - PA)$  because  $\mu \leq 1$ . If  $-PA \leq B < 0$  then  $h = 0$ . If  $B < -PA$  then  $h = 2\mu(|B| - PA)$ .  $\square$

**Lemma 11.** *Let  $q \geq 3$ ,  $b \geq 1$ ,  $N = T_b(q)$ ,  $F(w) = \|Nw\|_1$  and  $D = d_b(q)$ . Let  $P \geq 4$  and  $\frac{3}{5} \leq \mu \leq 1$ . For all  $y \in \{\pm 1\}^{b+1}$  and  $\zeta \in [-1, 1]^{b+1}$ ,*

$$\Lambda(y, \zeta) := P F(y - \zeta) + \mu F(\zeta + Py) - (P - \mu) F(\zeta) \leq P(1 + \mu) D,$$

*with strict inequality unless  $y = \pm s_b$ .*

*Proof.* Let  $\xi_j(w)$  be the path (1) with  $x = q$ ,  $k = b$ , and put  $\tau_{-1}(w) = \xi_{-1}(w)$ ,  $\tau_j(w) = \xi_{j-1}(w) - \xi_j(w)$  for  $0 \leq j < b$ . These are linear, and  $F(w) = q^b \sum_j |\tau_j(w)|$  by Lemma 7. With  $A_j = \tau_j(y)$  and  $B_j = \tau_j(\zeta)$ ,

$$\Lambda/q^b = P(1 + \mu)F(y)/q^b + \sum_j h(A_j, B_j).$$

By Lemma 8 (with  $x = q$  and potential  $\mu_j^q$ ), it suffices to show  $\sum_j h(A_j, B_j) \leq \sum_{j \in \text{NA}(y)} 2P(1 + \mu)\mu_j^q x_j$ , where  $x_j = |\xi_j(y)| \in [\frac{q-2}{q}, 1]$ , with strict inequality when  $\text{NA}(y) \neq \emptyset$ . Since  $\xi_j(\zeta)$  is a convex combination of entries of  $\zeta$ , we have  $|B_{-1}| \leq 1$  and  $|B_j| = \frac{q-1}{q} |\zeta_j - \xi_j(\zeta)| \leq \frac{2(q-1)}{q}$ . Also  $|A_{-1}| \geq \frac{q-2}{q}$ , and  $|A_j| = \frac{q-1}{q}(1 + x_j)$  for  $j \notin \text{NA}(y)$ ,  $|A_j| = \frac{q-1}{q}(1 - x_j)$  for  $j \in \text{NA}(y)$ . We use Lemma 10, noting  $P \geq 4 > \mu$ .

- $j = -1$ :  $P|A_{-1}| \geq \frac{4(q-2)}{q} \geq \frac{4}{3} > |B_{-1}|$ , so  $h \leq 0$ .
- $j \geq 0, j \notin \text{NA}(y)$ :  $P|A_j| \geq \frac{8(q-1)^2}{q^2} \geq \frac{2(q-1)}{q} \geq |B_j|$ , so  $h \leq 0$ .
- $j \in \text{NA}(y)$ : it suffices that  $\mu|B_j| < P\beta(x_j)$ , where  $\beta(t) = \mu \frac{q-1}{q}(1-t) + (1+\mu)\mu_j^q t$ . As  $\beta$  is affine it suffices to check the endpoints. For  $q = 3$ ,  $\beta(1) \geq \frac{8}{5} \cdot \frac{1}{3}$  and  $\beta(\frac{1}{3}) \geq \frac{3}{5} \cdot \frac{4}{9} + \frac{8}{5} \cdot \frac{1}{9} = \frac{4}{9}$ , so  $P\beta \geq \frac{16}{9} > \frac{4}{3} \geq \mu|B_j|$ . For general  $q \geq 3$  the same comparison reduces to  $P\beta(1) \geq \frac{32}{5} \cdot \frac{q-2}{q} > \frac{2(q-1)}{q}$  and  $\frac{22}{5}q^2 - \frac{94}{5}q + \frac{104}{5} > 0$ .

Summing over  $j$  proves the claim.  $\square$

## 6 The sign-matrix inequality

**Theorem 12.** *Let  $p \geq 5$  and  $q \geq 3$  be real,  $a, b \geq 1$ ,  $M = T_a(p)$  and  $N = T_b(q)$ . For every  $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$ ,*

$$\|MYN^\top\|_1 \leq d_a(p) d_b(q),$$

*with equality exactly when  $Y = \pm s_a s_b^\top$ .*

*Proof.* Let  $\bar{y}_0, \dots, \bar{y}_a$  be the rows of  $Y$ . Define  $\zeta_{a-1} = \bar{y}_a$  and  $\zeta_{k-1} = (\zeta_k + (p-1)\bar{y}_k)/p \in [-1, 1]^{b+1}$ . Column  $v$  of  $MYN^\top$  is  $Mw^{(v)}$  with  $w_i^{(v)} = (N\bar{y}_i)_v$ , and by linearity the path of  $w^{(v)}$  is  $((N\zeta_k)_v)_k$ . Applying Lemma 7 with  $x = p$  to every column and summing,

$$\|MYN^\top\|_1 = p^a \left( F(\zeta_{-1}) + \sum_{k=0}^{a-1} F(\zeta_{k-1} - \zeta_k) \right), \quad F(w) = \|Nw\|_1.$$

Let  $\mu_k$  be the potential of Lemma 8 for  $x = p$  and put  $f_k = F(\zeta_{k-1} - \zeta_k) + \mu_{k-1}F(\zeta_{k-1}) - \mu_k F(\zeta_k)$ . Telescoping gives  $\|MYN^\top\|_1/p^a = \sum_k f_k + \mu_{a-1}F(\bar{y}_a)$ . Since  $\zeta_{k-1} - \zeta_k = \frac{p-1}{p}(\bar{y}_k - \zeta_k)$  and  $p\mu_k = p-1 - \mu_{k-1}$ , we get  $pf_k = \Lambda(\bar{y}_k, \zeta_k)$  with  $P = p-1 \geq 4$  and  $\mu = \mu_{k-1} \in [\frac{p-2}{p}, 1] \subseteq [\frac{3}{5}, 1]$ . Lemma 11 gives  $f_k \leq c_k d_b(q)$ , and Lemma 8 gives  $F(\bar{y}_a) \leq d_b(q)$ ; both are strict unless the row involved is  $\pm s_b$ , and  $\mu_{a-1} > 0$ . Hence  $\|MYN^\top\|_1 \leq p^a(\sum_k c_k + \mu_{a-1})d_b(q) = d_a(p)d_b(q)$ . If equality holds, every row of  $Y$  is  $\pm s_b$ , so  $Y = gs_b^\top$  with  $g \in \{\pm 1\}^{a+1}$  and  $\|MYN^\top\|_1 = \|Mg\|_1 d_b(q)$ ; by Lemma 8 this equals  $d_a(p)d_b(q)$  only for  $g = \pm s_a$ .  $\square$

*Remark 13.* Since  $\|MYN^\top\|_1 = \|NY^\top M^\top\|_1$ , the roles of the two factors may be exchanged, so Theorem 12 holds whenever  $\min(p, q) \geq 3$  and  $\max(p, q) \geq 5$ . In particular it contains the sign-matrix inequality [6, Lemma 4.1] used for Theorem 1, and the deduction below then proves Theorem 1 as well. The semidefinite argument of [6] does not extend to  $x = 3$  [6, Remark 4.2]; the proof above instead uses the integrality of the signs on both factors.

## 7 Proof of Theorem 2

Let  $a = 2r$ ,  $b = 2s + 1$ ,  $M = T_a(p)$ ,  $N = T_b(3)$ , let  $X$  be the divisor matrix of  $\mathcal{D}$  (so  $x_{ab} = 0$ ), and let  $Y = 2X - J$  with  $J$  the all-ones matrix. By Proposition 6 and  $M\mathbf{1} = p^a e_a$ ,  $N\mathbf{1} = 3^b e_b$ ,

$$2\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \|MJN^\top + MYN^\top\|_1 = \|n e_a e_b^\top + MYN^\top\|_1 \leq n + d_a(p) d_b(3),$$

by Theorem 12. For the checkerboard set,  $Y^* = s_a s_b^\top$ ,  $x_{00}^* = 1$ , and  $x_{ab}^* = 0$  because  $a+b$  is odd, so  $\mathcal{D}_{r,s}^*$  is an admissible nonempty set. The  $(a, b)$  entry of  $MY^*N^\top = (Ms_a)(Ns_b)^\top$  is positive by Lemma 8, so it does not cancel against  $n$ , and  $2\mathcal{E}(\text{ICG}_n(\mathcal{D}_{r,s}^*)) = n + \|Ms_a\|_1 \|Ns_b\|_1 = n + d_a(p) d_b(3)$ . If  $\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \mathcal{E}(\text{ICG}_n(\mathcal{D}_{r,s}^*))$ , then  $\|MYN^\top\|_1 = d_a(p)d_b(3)$ , so  $Y = \pm s_a s_b^\top$  by Theorem 12; since  $y_{ab} = -1 = (s_a s_b^\top)_{ab}$ , we get  $Y = Y^*$  and  $\mathcal{D} = \mathcal{D}_{r,s}^*$ .  $\square$

## 8 Exponents of equal parity

Let  $p, q$  be distinct odd primes, let  $a, b \geq 1$  with  $a+b$  even,  $n = p^a q^b$ ,  $M = T_a(p)$  and  $N = T_b(q)$ . For a set  $\mathcal{D}$  of proper divisors of  $n$  let  $X$  and  $Y = 2X - J$  be as in Section 7, so that  $y_{ab} = -1$ , and let  $Z = MYN^\top$ , with entries  $Z_{ij}$ . The last rows of  $M$  and  $N$  are nonnegative with sums  $p^a$  and  $q^b$ , so  $|Z_{ab}| \leq n$ . Writing  $t^- = \max\{-t, 0\}$ , the computation of Section 7 gives

$$G(Y) := 2\mathcal{E}(\text{ICG}_n(\mathcal{D})) - n = \sum_{(i,j) \neq (a,b)} |Z_{ij}| + Z_{ab} = \|Z\|_1 - 2(Z_{ab})^-. \quad (2)$$

Since  $a+b$  is even, the sign matrix  $s_a s_b^\top$  has entry  $+1$  at  $(a, b)$  and violates the proper-divisor condition. The matrix  $-s_a s_b^\top$ , which corresponds to the set  $\{p^i q^j : i+j \text{ odd}\}$ , is admissible, and it attains the bound  $\|Z\|_1 \leq d_a(p) d_b(q)$  of Theorem 12. However, its entry  $Z_{ab} = -(Ms_a)_a (Ns_b)_b = -\delta_a(p) \delta_b(q)$  is negative by Remark 9, and it cancels against  $n$  in the argument of Section 7. So the energy bound  $\frac{1}{2}[n + d_a(p) d_b(q)]$  is not attained, and the maximiser changes. By (2), the set  $\{p^i q^j : i+j \text{ odd}\}$  has energy  $\frac{1}{2}[n + d_a(p) d_b(q)] - \delta_a(p) \delta_b(q)$ .

**Conjecture 14.** *Let  $p, q$  be distinct odd primes,  $a, b \geq 1$  with  $a+b$  even, and  $n = p^a q^b$ . The maximal energy of an integral circulant graph of order  $n$  is*

$$\frac{1}{2}[n + d_a(p) d_b(q)] - \delta_a(p) \delta_b(q),$$

*attained exactly by the sets  $\{p^i q^j : i+j \text{ odd}\}$  and  $\{p^i q^j : i+j \text{ even}\} \setminus \{n\}$ .*

By the computation above, the first of the two sets always attains the conjectured value. Since  $d_1(x) = 3x - 4$ ,  $\delta_1(x) = x - 2$ ,  $d_2(x) = 5x^2 - 8x + 4$  and  $\delta_2(x) = x^2 - 2x + 2$ , Theorems 3 and 4 are the cases  $\min\{a, b\} = 1$  and  $\min\{a, b\} = 2$  of the conjecture, and Theorem 5 is the case  $\min\{a, b\} \leq 8$ ; they are proved in Sections 9, 11 and 12. The next result is computer-assisted.

**Proposition 15.** *Conjecture 14 holds for the exponent pairs  $(a, b) = (2, 2), (2, 4), (4, 2), (2, 6), (6, 2), (2, 8), (8, 2), (3, 3), (3, 5), (5, 3)$  and  $(4, 4)$ , and all pairs of distinct odd primes  $p, q$ .*

*Proof (computer-assisted).* Exchanging the names of the primes transposes  $X$  and does not change the energy, and the list of exponent pairs is closed under transposition. As one of two distinct odd primes is at least 5, it therefore suffices to show, for each listed  $(a, b)$  and all real  $p \geq 5$ ,  $q \geq 3$ , that every  $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$  with  $y_{ab} = -1$  satisfies  $G(Y) \leq \Theta_{ab} := d_a(p)d_b(q) - 2\delta_a(p)\delta_b(q)$ , with equality exactly for  $Y = -s_a s_b^\top$  and  $Y = s_a s_b^\top - 2e_a e_b^\top$ , the sign matrices of the two sets in the conjecture; for real  $p, q$ ,  $G(Y)$  is defined by the middle expression in (2).

Substitute  $p = 5 + s$  and  $q = 3 + t$  with  $s, t \geq 0$ . Then every entry of  $Z$  is a polynomial in  $s$  and  $t$  with integer coefficients, and so is  $\Theta_{ab}$ , because  $d_k(x)$  and  $\delta_k(x) = x^k + 2 \sum_{i < k} (-1)^{k-i} x^i$  have integer coefficients (Remark 9). An entry whose coefficients are all  $\geq 0$ , or all  $\leq 0$ , has a known sign on the whole region. Choose a sign  $\varepsilon_{ij}$  for each of the other entries with  $(i, j) \neq (a, b)$ , and let  $L$  be the sum of  $Z_{ab}$ , of the entries of known sign multiplied by their signs, and of the  $\varepsilon_{ij} Z_{ij}$ . At every point of the region,  $G(Y)$  is the largest of these linear forms  $L$ . Hence, if for every choice of signs  $\Theta_{ab} - L$  either vanishes identically or has nonnegative coefficients and a positive constant term, then  $G(Y) \leq \Theta_{ab}$  on the region; equality holds identically if some choice gives  $\Theta_{ab} - L = 0$ , and  $G(Y) < \Theta_{ab}$  everywhere otherwise. For each listed  $(a, b)$ , a computer calculation in exact integer arithmetic over all  $2^{(a+1)(b+1)-1}$  admissible sign matrices (up to  $2^{26}$ ) shows that this condition holds for every  $Y$ , and that some choice gives  $\Theta_{ab} - L = 0$  exactly when  $Y$  is one of the two matrices above. The programs are described in Section 13.  $\square$

Every exponent pair in Proposition 15 has  $\min\{a, b\} \leq 4$ , so the proposition is contained in Theorem 5, and for the pairs  $(2, b)$  and  $(b, 2)$  in Theorem 4. Its proof treats each sign matrix separately and does not use Sections 10 to 12, so it is an independent check of these cases. Conjecture 14 remains open when  $\min\{a, b\} \geq 9$ ; the smallest open exponent pairs are  $(9, 9)$ ,  $(9, 11)$ ,  $(11, 9)$  and  $(10, 10)$ .

*Remark 16.* For  $n = pq$ , Theorem 3 gives the maximal energy  $4(p-1)(q-1)$ , attained exactly by  $\{p, q\}$  and  $\{1\}$ . This can also be read off from known formulas. The graph  $\text{ICG}_{pq}(\{1\})$  is the unitary Cayley graph, whose energy  $4\varphi(pq)$  was computed by Ilić [3], and  $\mathcal{E}(\text{ICG}_{pq}(\{p, q\})) = 4\varphi(pq)$  by [4, Theorem 4.2]. The graphs  $\text{ICG}_{pq}(\{p\})$  and  $\text{ICG}_{pq}(\{1, p\})$  are  $p$  disjoint copies of  $K_q$  and the complete  $q$ -partite graph with parts of size  $p$  (see also [4, Theorem 4.1]); both have energy  $2p(q-1)$ , and symmetrically for  $\{q\}$  and  $\{1, q\}$ . Finally  $\text{ICG}_{pq}(\{1, p, q\})$  is the complete graph  $K_{pq}$ , of energy  $2(pq-1)$ . We have not found the maximal energy for  $n = pq$  stated explicitly in the literature. Le and Sander [5, Theorem 2.1 and Corollary 2.1] determined the maximum  $\tilde{E}_{\max}(n)$  of the energy over *multiplicative* divisor sets and showed  $E_{\max}(n) \geq \tilde{E}_{\max}(n)$ . For  $n = pq$  equality holds, but for  $n = pq^3$  it fails: by direct computation,  $E_{\max}(5 \cdot 3^3) = 584 > 512 = \tilde{E}_{\max}(5 \cdot 3^3)$  and  $E_{\max}(3 \cdot 5^3) = 1632 > 1472 = \tilde{E}_{\max}(3 \cdot 5^3)$ , and in both cases the maximisers are the non-multiplicative sets of Theorem 3.

*Remark 17.* Other sets come close to the maximum. Let  $(a, b) = (2, 2)$ ,  $q = 3$  and  $\mathcal{D} = \{1, 3, p^2, 3p^2, 9p\}$ , so that  $Y = s_2 y^\top$  with  $y = (1, 1, -1)^\top$  and  $Z = (M_{s_2})(Ny)^\top$ . The path of  $y$  is  $z_1 = -1$ ,  $z_0 = \frac{1}{3}$ ,  $z_{-1} = \frac{7}{9}$ , and  $\text{NA}(y) = \{0\}$ , so Lemma 8 gives  $\|Ny\|_1 = d_2(3) - 2 = 23$ , and  $Z_{22} = 9\delta_2(p)z_{-1} > 0$ . By (2),  $G(Y) = 23d_2(p)$ , while the maximum of  $G$  is  $25d_2(p) - 10\delta_2(p) = G(Y) + 4(p-3)$  (Theorem 25). So the energy of  $\mathcal{D}$  is smaller than the maximal energy  $62p^2 - 90p + 40$  only by  $2(p-3)$ .

## 9 Proof of Theorem 3

Let  $p, q, m$  and  $n = pq^m$  be as in Theorem 3; the case  $n = p^m q$  follows by exchanging the names of the primes. Put  $x = q$ ,  $P = x - 1$  and  $R = p - 1$ . As  $p$  and  $q$  are distinct odd primes, either  $x \geq 5$  and  $R \geq 2$ , or  $x = 3$  and  $R \geq 4$ ; only these inequalities are used. In both cases

$$R + 1 < P(R - 1). \quad (3)$$

Let  $\mu_j$  and  $c_j$  be the potential of Lemma 8 for  $x$ , and put  $\rho = \mu_{m-1}$ ,  $\hat{c}_j = c_j$  for  $0 \leq j < m$ ,  $\hat{c}_m = \rho$  and  $\hat{D} = \sum_{j=0}^m \hat{c}_j$ . By Lemma 8 and Remark 9,  $\hat{D} = d_m(x)/x^m$  and  $\rho = \delta_m(x)/x^m$ . For  $w \in \mathbb{R}^{m+1}$  let  $z_j(w)$ ,  $-1 \leq j \leq m-1$ , be the path (1) with  $k = m$ , and let  $\nu(w) = \|T_m(x)w\|_1/x^m$ . For a sign vector  $y$  put  $\sigma(y) = \sum_{j \in \text{NA}(y)} \mu_j |z_j(y)|$ , so that  $\nu(y) = \hat{D} - 2\sigma(y)$  by Lemma 8. For a statement  $S$ ,  $[S]$  is 1 if  $S$  holds and 0 otherwise.

For  $0 \leq \mu \leq 1$  let  $h_\mu$  be the function  $h$  of Lemma 10 with  $P = x - 1$ . Besides Lemma 10 we use three values, which follow from the definition and from the case analysis in the proof of Lemma 10:  $h_\mu(0, B) = 2\mu|B|$ ;  $h_\mu(A, B) = -2(P - \mu)|B|$  if  $AB > 0$  and  $|B| \leq |A|$ ; and  $h_\mu(A, B) = 0$  if  $AB \leq 0$  and  $|B| \leq P|A|$ .

Let  $\mu_\infty = P/(x+1)$ . As  $x\mu_\infty = P - \mu_\infty$ , the recursion for  $\mu$  gives  $\mu_j - \mu_\infty = (-1/x)^{j+1}(1 - \mu_\infty)$ , where  $1 - \mu_\infty > 0$ . Hence, for  $j \geq 0$ ,

$$\mu_j > \mu_\infty \text{ if } j \text{ is odd,} \quad \mu_j < \mu_\infty \text{ if } j \text{ is even,} \quad \mu_0 \leq \mu_j \leq \mu_1. \quad (4)$$

In particular  $\rho < \mu_\infty$ , because  $m - 1$  is even.

**Lemma 18.** For every  $w \in \mathbb{R}^{m+1}$ ,

$$\nu(w) = \sum_{j=0}^m \hat{c}_j |w_j| + \frac{1}{x} \sum_{j=0}^{m-1} h_{\mu_{j-1}}(w_j, z_j(w)).$$

*Proof.* Write  $z_j = z_j(w)$ . By (1),  $z_{j-1} - z_j = \frac{P}{x}(w_j - z_j)$  and  $xz_{j-1} = z_j + Pw_j$ . Together with  $x\mu_j = P - \mu_{j-1}$  this gives, for  $0 \leq j < m$ ,

$$|z_{j-1} - z_j| + \mu_{j-1}|z_{j-1}| - \mu_j|z_j| = \frac{1}{x} [h_{\mu_{j-1}}(w_j, z_j) + P(1 + \mu_{j-1})|w_j|].$$

Summing over  $j$ , the potential terms telescope to  $\mu_{-1}|z_{-1}| - \mu_{m-1}|z_{m-1}| = |z_{-1}| - \rho|w_m|$ , and Lemma 7 gives the claim.  $\square$

**Lemma 19.** For  $y \in \{\pm 1\}^{m+1}$  we have  $\nu(y) - |z_{-1}(y)| \leq \hat{D} - \rho$ , with equality exactly when  $y = \pm s_m$ .

*Proof.* By Lemma 7, the left side is  $\sum_{j < m} |z_{j-1} - z_j|$ , where  $z_j = z_j(y)$ . Let  $\mu'_{-1} = 0$  and  $\mu'_j = (P - \mu'_{j-1})/x$  for  $j \geq 0$ , so that  $\mu'_j \in (0, P/x]$  for  $j \geq 0$ . The computation in the proof of Lemma 8 uses only this recursion of the potential, so it gives

$$|z_{j-1} - z_j| + \mu'_{j-1}|z_{j-1}| - \mu'_j|z_j| = c'_j - 2\mu'_j|z_j| \quad [j \in \text{NA}(y)], \quad c'_j = P(1 + \mu'_{j-1})/x.$$

Summing over  $j < m$  and using  $\mu'_{-1} = 0$  and  $|z_{m-1}| = 1$ ,

$$\sum_{j < m} |z_{j-1} - z_j| = \sum_{j < m} c'_j + \mu'_{m-1} - 2 \sum_{j \in \text{NA}(y)} \mu'_j |z_j|.$$

All terms of the last sum are positive, so the left side is maximal exactly when  $\text{NA}(y) = \emptyset$ , that is, when  $y = \pm s_m$ . For  $y = s_m$  it equals  $\nu(s_m) - |z_{-1}(s_m)| = \hat{D} - \rho$ , by Lemma 8 and Remark 9.  $\square$

**Lemma 20.** *Let  $y \in \{\pm 1\}^{m+1}$  with  $y_0 = y_m$ . Then  $(3R - 1)\sigma(y) > (R - 1)\rho$ .*

*Proof.* As  $m$  is odd and  $y_0 = y_m$ ,  $y \neq \pm s_m$ , so  $\text{NA}(y)$  is nonempty; let  $J = \max \text{NA}(y)$ . Since  $y$  alternates on  $J + 1, \dots, m$ , the recursion  $|z_{j-1}| = (P - |z_j|)/x$  from the proof of Lemma 8 holds for  $J < j \leq m - 1$ , and starting from  $|z_{m-1}| = 1 = \mu_{-1}$  it gives  $|z_J(y)| = \mu_{m-2-J}$ . Hence  $\sigma(y) \geq \mu_J \mu_{m-2-J}$ . If  $J = m - 1$ , this is  $\rho$ , and  $(3R - 1)\rho > (R - 1)\rho$ . Otherwise  $J$  and  $m - 2 - J$  are nonnegative and have opposite parity, since their sum is odd, so  $\sigma(y) > \mu_0 \mu_\infty$  by (4). As  $(3R - 1)(x - 2) - (R - 1)x = 2R(x - 3) + 2 > 0$ , we have  $(3R - 1)\mu_0 > R - 1$ , and therefore  $(3R - 1)\sigma(y) > (3R - 1)\mu_0 \mu_\infty \geq (R - 1)\mu_\infty \geq (R - 1)\rho$ .  $\square$

*Proof of Theorem 3.* Let  $\mathcal{D}$  be a set of proper divisors of  $n$ , and let  $u$  and  $v$  be the two rows of its sign matrix  $Y$ , so that  $v_m = -1$ . Since  $T_1(p) = \begin{pmatrix} -R & R \\ R & 1 \end{pmatrix}$ , the rows of  $Z = T_1(p)YT_m(x)^\top$  are  $-RT_m(x)w^1$  and  $T_m(x)w^2$ , where  $w^1 = u - v$  and  $w^2 = Ru + v$ . By the proof of Lemma 7,  $Z_{1m} = x^m z_{-1}(w^2)$ , so (2) gives  $G(Y) = x^m \Phi$  with

$$\Phi = R\nu(w^1) + \nu(w^2) - 2(z_{-1}(w^2))^-.$$

As  $d_1(p) = 3R - 1$  and  $\delta_1(p) = R - 1$ , the claimed maximal energy is  $\frac{1}{2}(n + x^m \Theta)$  with  $\Theta = (3R - 1)\hat{D} - 2(R - 1)\rho$ . We show that  $\Phi \leq \Theta$ , with equality exactly for  $Y = s_1 s_m^\top - 2e_1 e_m^\top$  and  $Y = -s_1 s_m^\top$ . These are the sign matrices of the two sets in the theorem, which are nonempty sets of proper divisors of  $n$  because  $1 + m$  is even; so the theorem follows.

Let  $\mathcal{A} = \{j : u_j = v_j\}$ . For  $j \in \mathcal{A}$  we have  $w_j^1 = 0$  and  $|w_j^2| = R + 1$ ; for  $j \notin \mathcal{A}$  we have  $|w_j^1| = 2$  and  $|w_j^2| = R - 1$ . So  $R|w_j^1| + |w_j^2|$  is  $R + 1$  on  $\mathcal{A}$  and  $3R - 1$  off  $\mathcal{A}$ , and Lemma 18, applied to  $w^1$  and  $w^2$ , gives

$$\Theta - \Phi = 2(R - 1) \left( \sum_{j \in \mathcal{A}} \hat{c}_j - \rho \right) - H + 2(z_{-1}(w^2))^- , \quad (5)$$

where  $H$  is the sum of the  $2m$  cells  $\frac{R}{x} h_{\mu_{j-1}}(w_j^1, z_j(w^1))$  and  $\frac{1}{x} h_{\mu_{j-1}}(w_j^2, z_j(w^2))$ ,  $0 \leq j < m$ . Recall that  $z_j(w)$  is a convex combination of  $w_{j+1}, \dots, w_m$ .

- (a) For the  $w^2$ -cells,  $|z_j(w^2)| \leq R + 1 < P(R - 1) \leq P|w_j^2|$  by (3), so they are  $\leq 0$  by Lemma 10.
- (b) For the  $w^1$ -cells with  $j \notin \mathcal{A}$ ,  $|z_j(w^1)| \leq 2 \leq P|w_j^1|$ , so they are  $\leq 0$ .
- (c) A  $w^1$ -cell with  $j \in \mathcal{A}$  equals  $\frac{R}{x} h_{\mu_{j-1}}(0, z_j(w^1)) = \frac{2R}{x} \mu_{j-1} |z_j(w^1)|$ .

For  $j \in \mathcal{A}$  with  $j < m$  let

$$s_j = 2(R - 1)c_j - \frac{2R}{x} \mu_{j-1} |z_j(w^1)| \geq \frac{2}{x} [(R - 1)P(1 + \mu) - 2R\mu], \quad \mu = \mu_{j-1},$$

where we used  $|z_j(w^1)| \leq 2$ . The bracket is affine in  $\mu$ , and it equals  $(R - 1)P > 0$  at  $\mu = 0$  and  $2[R(P - 1) - P] > 0$  at  $\mu = 1$ ; hence  $s_j > 0$ . Let  $S = \sum_{j \in \mathcal{A}, j < m} s_j$ , let  $C \geq 0$  be minus the sum of the cells in (a) and (b), and let  $K = 2(z_{-1}(w^2))^- \geq 0$ . Then (5) reads

$$\Theta - \Phi = S + C + K - 2(R - 1)\rho [m \notin \mathcal{A}]. \quad (6)$$

Moreover  $|z_0(w^2)| \leq R + 1 < P|w_0^2|$ , so  $z_{-1}(w^2) = (z_0(w^2) + Pw_0^2)/x$  has the sign of  $w_0^2 = Ru_0 + v_0$ , which is the sign of  $u_0$ . Thus  $K = 0$  if and only if  $u_0 = 1$ .

*Case 1:  $m \in \mathcal{A}$ .* Then  $\Theta - \Phi = S + C + K \geq 0$ . Suppose that equality holds. Then  $S = 0$ , so  $\mathcal{A} = \{m\}$ ;  $K = 0$ , so  $u_0 = 1$ ; and  $C = 0$ . Now  $w_j^1 = 2u_j$  for  $j < m$  and  $w_m^1 = 0$ . For  $j \leq m - 2$  we have  $|Pw_{j+1}^1| = 2P > 2 \geq |z_{j+1}(w^1)|$ , so  $z_j(w^1) = (z_{j+1}(w^1) + Pw_{j+1}^1)/x$  has the sign of  $u_{j+1}$ , and  $|z_j(w^1)| \leq 2 = |w_j^1|$ . If  $u_j = u_{j+1}$  for some  $j \leq m - 2$ , the  $w^1$ -cell at  $j$  is

therefore  $-\frac{2R}{x}(P - \mu_{j-1})|z_j(w^1)| < 0$ , contradicting  $C = 0$ . So  $u$  alternates on  $0, \dots, m-1$  and  $u_0 = 1$ ; together with  $u_m = v_m = -1$  and  $v_j = -u_j$  for  $j < m$ , this means  $Y = s_1 s_m^\top - 2e_1 e_m^\top$ , as  $m$  is odd. Conversely, for this  $Y$  we have  $\mathcal{A} = \{m\}$ ,  $u = s_m$  and  $S = K = 0$ . The  $w^1$ -cells vanish: for  $j \leq m-2$ ,  $w_j^1 = 2u_j$  and  $z_j(w^1)$  have opposite signs and  $|z_j(w^1)| \leq P|w_j^1|$ , and  $z_{m-1}(w^1) = w_m^1 = 0$ . The  $w^2$ -cells vanish too:  $w_i^2$  has the sign of  $u_i$  for every  $i$ ; by (3),  $z_j(w^2) = (z_{j+1}(w^2) + Pw_{j+1}^2)/x$  has the sign of  $w_{j+1}^2$  for  $j \leq m-2$ , and  $z_{m-1}(w^2) = w_m^2$ ; since  $u$  alternates,  $w_j^2$  and  $z_j(w^2)$  have opposite signs, and  $|z_j(w^2)| \leq P|w_j^2|$  by (a). Hence  $\Phi = \Theta$ .

*Case 2:*  $m \notin \mathcal{A} \neq \emptyset$ . Then  $u_m = 1$  and  $\mathcal{A} \subseteq \{0, \dots, m-1\}$ . By (6) it suffices to find  $k \in \mathcal{A}$  with  $s_k > 2(R-1)\rho$ . Since  $\rho < \mu_\infty = P/(P+2)$ , the lower bound for  $s_k$  shows that this holds if

$$(R-1)P + \mu(P+2)[R(P-2) - P] \geq 0, \quad \mu = \mu_{k-1}. \quad (7)$$

If  $x \geq 5$ , then  $P \geq 4$  and  $R(P-2) - P \geq P-4 \geq 0$ , so (7) holds for every  $k \in \mathcal{A}$ . If  $x = 3$ , then (7) reads  $2(R-1) \geq 8\mu$ . If some  $k \in \mathcal{A}$  has  $k \geq 1$ , then  $\mu_{k-1} \leq \mu_1 = \frac{5}{9}$  by (4), and  $2(R-1) \geq 6 > \frac{40}{9}$ . It remains to treat  $x = 3$  and  $\mathcal{A} = \{0\}$ . Here  $\mu_\infty = \frac{1}{2}$  and  $c_0 = \frac{4}{3}$ , and since  $z_0(w)$  depends only on  $w_1, \dots, w_m$ , we have  $z_0(w^1) = 2z_0(u)$  and  $s_0 = \frac{2}{3}[4(R-1) - 2R|z_0(u)|]$ . If  $m = 1$ , then  $|z_0(u)| = 1$ ,  $\rho = \mu_0 = \frac{1}{3}$  and  $s_0 = \frac{2}{3}(2R-4) > \frac{2}{3}(R-1) = 2(R-1)\rho$ . Let  $m \geq 3$ . If  $|z_0(u)| \leq \frac{5(R-1)}{4R}$ , then  $s_0 \geq R-1 > 2(R-1)\rho$ . Otherwise  $|z_0(u)| > \frac{15}{16}$ . If  $u_1 \neq u_2$ , then  $u_1 = -\text{sgn } z_1(u)$  by Lemma 8, and  $|z_0(u)| = (2 - |z_1(u)|)/3 \leq \frac{5}{9}$ ; hence  $u_1 = u_2$ . Then  $1 \notin \mathcal{A}$ , and the arguments  $w_1^1 = 2u_1$  and  $z_1(w^1) = 2z_1(u)$  of the  $w^1$ -cell at  $j = 1$  have the same sign, with  $|z_1(w^1)| \leq |w_1^1|$ . This cell equals  $-\frac{2R}{3}(P - \mu_0)|z_1(w^1)| = -\frac{20R}{9}|z_1(u)| \leq -\frac{20R}{27}$ , so  $C \geq \frac{20R}{27}$ . With  $s_0 \geq \frac{2}{3}(2R-4)$  and  $2(R-1)\rho < R-1$ ,

$$\Theta - \Phi \geq \frac{2}{3}(2R-4) + \frac{20R}{27} - 2(R-1)\rho > \frac{56R-72}{27} - (R-1) = \frac{29R-45}{27} > 0.$$

*Case 3:*  $\mathcal{A} = \emptyset$ , that is,  $v = -u$ . Then  $u_m = 1$ ,  $w^1 = 2u$ ,  $w^2 = (R-1)u$  and  $\Phi = (3R-1)\nu(u) - 2(R-1)(z_{-1}(u))^-$ . If  $u_0 = 1$ , then  $z_{-1}(u) > 0$  by Lemma 8, and  $\Phi = (3R-1)(\hat{D} - 2\sigma(u)) < \Theta$  by Lemma 20, since  $u_0 = u_m$ . If  $u_0 = -1$ , then  $(z_{-1}(u))^- = |z_{-1}(u)|$  and

$$\Phi = (R+1)\nu(u) + 2(R-1)(\nu(u) - |z_{-1}(u)|) \leq (R+1)\hat{D} + 2(R-1)(\hat{D} - \rho) = \Theta$$

by Lemmas 8 and 19, with equality exactly when  $u = \pm s_m$ , that is,  $u = -s_m$ . This is  $Y = -s_1 s_m^\top$ .

The three cases cover all  $Y$ , which proves the theorem. For  $m = 1$  the value is  $4(p-1)(q-1)$ , since  $d_1(q) = 3q-4$  and  $\delta_1(q) = q-2$ .  $\square$

## 10 The column path

In this section and the next two, the exponent  $a \geq 2$  of  $p$  is fixed,  $b \geq 1$  is any integer with  $a+b$  even, and  $p, q$  are real numbers with

$$p \geq 5 \text{ and } q \geq 3, \quad \text{or} \quad p = 3 \text{ and } q \geq 5. \quad (8)$$

Every ordered pair of distinct odd primes satisfies (8). We use the notation of Section 9 with  $m$  replaced by  $b$ :  $x = q$ ,  $P = x-1$  and  $R = p-1$ ;  $\mu_j$  and  $c_j$  form the potential of Lemma 8 for  $x$ , and  $\mu_\infty = P/(x+1)$ ;  $\rho = \mu_{b-1}$ ,  $\hat{c}_j = c_j$  for  $j < b$ ,  $\hat{c}_b = \rho$  and  $\hat{D} = \sum_{j=0}^b \hat{c}_j$ , so that  $\hat{D} = d_b(q)/x^b$  and  $\rho = \delta_b(q)/x^b$ ; for  $w \in \mathbb{R}^{b+1}$ ,  $z_j(w)$  is the path (1) with  $k = b$  and  $\nu(w) = \|T_b(x)w\|_1/x^b$ ; and  $h_\mu$  is the function  $h$  of Lemma 10 with  $P = x-1$ . Lemma 18 and (4) hold with the same proofs, which do not use that  $m$  is odd. By Lemma 8 and (4),

$$\mu_0 \leq \mu_j \leq 1 \quad (j \geq -1), \quad \mu_j \leq \mu_1 \quad (j \geq 0), \quad \mu_0 = \frac{P-1}{x}, \quad \mu_1 = \frac{P^2+1}{x^2}. \quad (9)$$

By (8), either  $R \geq 4$  and  $P \geq 2$ , or  $R = 2$  and  $P \geq 4$ ; in particular  $P \geq 2$  and  $PR \geq 8$ .

**Values of the cell function.** Besides the three values of  $h_\mu$  listed in Section 9, the case analysis in the proof of Lemma 10, together with  $h_\mu(-A, -B) = h_\mu(A, B)$ , gives

$$h_\mu(A, B) = \begin{cases} 2(\mu|B| - P|A|) & \text{if } AB > 0 \text{ and } |B| \geq |A|, \\ 2\mu(|B| - P|A|) & \text{if } AB \leq 0 \text{ and } |B| \geq P|A|. \end{cases} \quad (10)$$

Consequently, for  $0 \leq \mu \leq 1$ ,

$$h_\mu(A, B) < 0 \quad \text{if } AB > 0 \text{ and } |B| < P|A|, \quad (11)$$

since the value is then  $-2(P - \mu)|B|$  or  $2(\mu|B| - P|A|) \leq 2(|B| - P|A|)$ . The assumption  $|B| < P|A|$  is needed here:  $h_\mu(A, PA/\mu) = 0$  for  $A > 0$  and  $0 < \mu \leq 1$ .

**The matrix  $M$ .** Let  $M = T_a(p)$ ,  $\alpha = Ms_a$  and  $u = Me_a$ , and let  $\omega_i$  be the sum of the absolute values of the entries of row  $i$  of  $M$ . Thus  $\|\alpha\|_1 = d_a(p)$ ,  $\alpha_a = \delta_a(p)$  by Remark 9, and  $|(M\zeta)_i| \leq \omega_i$  for  $\zeta \in [-1, 1]^{a+1}$ . Let  $\mu_j^{(p)}$  denote the potential of Lemma 8 for  $p$  in place of  $x$ . For  $i < a$ ,

$$u_i = p^{a-1-i}R, \quad \omega_i = 2p^{a-1}R, \quad \alpha_i = (-1)^{a-i}p^{a-1}R(1 + \mu_{i-1}^{(p)}), \quad (12)$$

and  $u_a = 1$ ,  $\omega_a = p^a$ . Indeed, by the proof of Lemma 7 with  $x = p$  and  $k = a$ , row  $i < a$  of  $M$  has the entry  $-p^{a-1}R$  in column  $l = a - 1 - i$  and nonnegative entries with sum  $p^{a-1}R$  in the columns after  $l$ , and  $\alpha_i = p^{a-1}R(z_l - (s_a)_l)$  for the path  $z$  of  $s_a$ , where  $\text{sgn } z_l = (s_a)_{l+1}$  and  $|z_l| = \mu_{a-2-l}^{(p)}$  by Lemma 8 and Remark 9; the last row of  $M$  is  $(p^{a-1}R, \dots, pR, R, 1)$ . Moreover  $\delta_a(p) = p^a \mu_{a-1}^{(p)} \geq p^{a-1}(p - 2) \geq 3$ . Hence

$$2u_i \leq |\alpha_i| \quad (i < a), \quad \delta_a(p) \geq 3, \quad P\delta_a(p) > p^a, \quad \omega_i \leq P|\alpha_i| \quad (0 \leq i \leq a). \quad (13)$$

The first inequality reads  $2 \leq p^i(1 + \mu_{i-1}^{(p)})$ , with equality for  $i = 0$ . Next,  $P\delta_a(p) \geq 2p^{a-1}(p - 2) > p^a$  if  $p \geq 5$ , and  $P\delta_a(p) \geq 4 \cdot 3^{a-1} > 3^a$  if  $p = 3$ . Finally  $\omega_i \leq 2|\alpha_i| \leq P|\alpha_i|$  for  $i < a$ , and  $\omega_a = p^a < P\delta_a(p)$ .

**Columns.** For  $c \in \{\pm 1\}^{a+1}$  put  $\Delta(c) = d_a(p) - \|Mc\|_1$ . By Lemma 8,  $\Delta(c) \geq 0$ , with equality exactly for  $c = \pm s_a$ . Let  $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$  have columns  $c_0, \dots, c_b$ . Define  $\zeta_{b-1} = c_b$  and  $\zeta_{j-1} = (\zeta_j + Pc_j)/x$  for  $j = b-1, \dots, 0$ ; this is the path (1) applied to the sequence of columns. Each  $\zeta_j$  is a convex combination of  $c_{j+1}, \dots, c_b$ , so  $\zeta_j \in [-1, 1]^{a+1}$ , and if  $r_0, \dots, r_a$  are the rows of  $MY$ , then  $z_j(r_i) = (M\zeta_j)_i$  by linearity. For  $0 \leq \mu \leq 1$ ,  $c \in \{\pm 1\}^{a+1}$  and  $\zeta \in [-1, 1]^{a+1}$  let

$$\psi_\mu(c, \zeta) = P(1 + \mu) \Delta(c) - \sum_{i=0}^a h_\mu((Mc)_i, (M\zeta)_i);$$

we call the terms  $h_\mu((Mc)_i, (M\zeta)_i)$  the *cells* of  $\psi_\mu(c, \zeta)$ . As  $h_\mu(-A, -B) = h_\mu(A, B)$ , we have  $\psi_\mu(-c, -\zeta) = \psi_\mu(c, \zeta)$ . For  $0 \leq j < b$  put  $\psi_j = \psi_{\mu_{j-1}^{(p)}}(c_j, \zeta_j)$ .

Let  $N = T_b(q)$  and  $\Theta = d_a(p) d_b(q) - 2\delta_a(p) \delta_b(q)$ , and let  $G(Y)$  be the middle expression in (2) with  $Z = MYN^\top$ . Let  $Y^- = -s_a s_b^\top$  and  $Y^+ = s_a s_b^\top - 2e_a e_b^\top$ . The columns of  $Y^-$  are  $(-1)^{j+1} s_a$ ; those of  $Y^+$  are  $(-1)^j s_a$  for  $j < b$  and  $g_+ = (-1)^a s_a - 2e_a$  for  $j = b$ , as  $(-1)^b = (-1)^a$ . By Section 8,  $G(Y^-) = \Theta$ .

**Lemma 21.** For every  $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$ ,

$$x^{1-b}(\Theta - G(Y)) = \sum_{j=0}^{b-1} \psi_j + x\rho(\Delta(c_b) - 2\delta_a(p)) + 2x((M\zeta_{-1})_a)^-.$$

*Proof.* Row  $i$  of  $Z$  is  $(Nr_i)^\top$ , so the absolute values of its entries have sum  $x^b \nu(r_i)$ , and  $Z_{ab} = (Nr_a)_b = x^b z_{-1}(r_a) = x^b (M\zeta_{-1})_a$  by the proof of Lemma 7. Hence  $G(Y) = x^b [\sum_i \nu(r_i) - 2((M\zeta_{-1})_a)^-]$  by (2). We apply Lemma 18 to each  $r_i$  and add. As  $\sum_i |(r_i)_j| = \|Mc_j\|_1 = d_a(p) - \Delta(c_j)$ ,  $z_j(r_i) = (M\zeta_j)_i$  and  $\hat{c}_j = P(1 + \mu_{j-1})/x$  for  $j < b$ , this gives

$$x^{-b}G(Y) = d_a(p)\hat{D} - \sum_{j=0}^b \hat{c}_j \Delta(c_j) + \frac{1}{x} \sum_{j=0}^{b-1} \sum_{i=0}^a h_{\mu_{j-1}}((Mc_j)_i, (M\zeta_j)_i) - 2((M\zeta_{-1})_a)^-.$$

Since  $x^{-b}\Theta = d_a(p)\hat{D} - 2\delta_a(p)\rho$  and  $\hat{c}_b = \rho$ , subtracting and multiplying by  $x$  proves the lemma.  $\square$

**Lemma 22.** *Let  $0 \leq \mu \leq 1$ ,  $c \in \{\pm 1\}^{a+1}$  and  $\zeta \in [-1, 1]^{a+1}$ . Then*

$$\psi_\mu(c, \zeta) \geq P(1 + \mu)\Delta(c) - 2\mu \sum_{i=0}^a \max\{0, |(M\zeta)_i| - P|(Mc)_i|\}.$$

*If  $c = \pm s_a$ , all cells of  $\psi_\mu(c, \zeta)$  are  $\leq 0$ ; so  $\psi_\mu(c, \zeta) \geq 0$ , with equality exactly when all cells vanish.*

*Proof.* The bound follows from Lemma 10, which applies as  $P \geq 2 > \mu$ . If  $c = \pm s_a$ , then  $\Delta(c) = 0$  and  $|(M\zeta)_i| \leq \omega_i \leq P|\alpha_i| = P|(Mc)_i|$  by (13), so every cell is  $\leq 0$  by Lemma 10.  $\square$

**Lemma 23.** *Let  $g \in \{\pm 1\}^{a+1}$  and  $0 \leq k \leq b$ , and suppose that  $c_b = g$  and  $c_{b-i} = (-1)^i g$  for  $1 \leq i \leq k$ . Then  $\zeta_{b-1-i} = (-1)^i \mu_{i-1} g$  for  $0 \leq i \leq k$ , and  $\psi_{b-i} = P(1 + \mu_{b-i-1})\Delta(g)$  for  $1 \leq i \leq k$ . In particular, if  $c_l = (-1)^{l+1} s_a$  for  $J < l \leq b$ , then  $\zeta_J = (-1)^J \mu_{b-2-J} s_a$ .*

*Proof.* The formula for  $\zeta_{b-1-i}$  holds for  $i = 0$  as  $\mu_{-1} = 1$ , and if it holds for  $i - 1$ , then  $\zeta_{b-1-i} = ((-1)^{i-1} \mu_{i-2} g + P(-1)^i g)/x = (-1)^i \mu_{i-1} g$ . For  $1 \leq i \leq k$ , the cells of  $\psi_{b-i}$  have the arguments  $A = (-1)^i M g$  and  $B = M \zeta_{b-i} = -\mu_{i-2} A$ , so that  $A_l B_l \leq 0$  and  $|B_l| \leq |A_l| \leq P|A_l|$  for every  $l$ , and all cells vanish. For the last claim take  $g = (-1)^{b+1} s_a$  and  $i = b - 1 - J$ .  $\square$

**Lemma 24.** *We have  $\Delta(g_+) = 2\delta_a(p)$ . For  $Y = Y^+$ , every  $\psi_j$  vanishes and  $(M\zeta_{-1})_a > 0$ , so that  $G(Y^+) = \Theta$ . Conversely, if  $c_b = g_+$  and, for every  $j < b$ ,  $c_j = \pm s_a$  and  $\psi_j = 0$ , then  $Y = Y^+$ .*

*Proof.* By (12),  $(Mg_+)_i = (-1)^a \alpha_i - 2u_i = (-1)^i (|\alpha_i| - 2(-1)^i u_i)$ , and  $|\alpha_i| - 2(-1)^i u_i \geq 0$  by (13). Hence  $\|Mg_+\|_1 = \sum_i |\alpha_i| - 2 \sum_i (-1)^i u_i = d_a(p) - 2s_a^\top M e_a = d_a(p) - 2\delta_a(p)$ , as  $M$  is symmetric.

Let  $Y = Y^+$ . By induction on  $j$  downwards,  $\zeta_j = (-1)^{j+1} \mu_{b-2-j} s_a - 2x^{j+1-b} e_a$  for  $-1 \leq j \leq b-1$ . At a column  $j < b$ , the cells of  $\psi_j$  have the arguments  $A_i = (-1)^j \alpha_i$  and  $B_i = (M\zeta_j)_i$ , and with  $\sigma_i = \text{sgn } A_i = (-1)^{j+a-i}$ ,

$$\sigma_i B_i = -\mu_{b-2-j} |\alpha_i| - 2(-1)^{j+a-i} x^{j+1-b} u_i.$$

We claim that  $-P|A_i| \leq \sigma_i B_i \leq 0$ , so that every cell vanishes. The upper bound holds if  $\mu_{b-2-j} |\alpha_i| \geq 2x^{j+1-b} u_i$ : for  $j = b-1$  this is  $|\alpha_i| \geq 2u_i$ , and for  $j \leq b-2$  it follows from  $\mu_{b-2-j} \geq \mu_0 = (P-1)/x$ ,  $x^{j+1-b} \leq 1/x$  and  $(P-1)|\alpha_i| \geq |\alpha_i| \geq 2u_i$ , where  $|\alpha_a| = \delta_a(p)$  and  $u_a = 1$ . The lower bound holds as  $\mu_{b-2-j} |\alpha_i| + 2u_i \leq 2|\alpha_i| \leq P|\alpha_i|$ . Since  $\Delta(\pm s_a) = 0$ , every  $\psi_j$  vanishes. Moreover  $(M\zeta_{-1})_a = \mu_{b-1} \delta_a(p) - 2x^{-b} \geq ((P-1)\delta_a(p) - 2)/x > 0$ . By Lemma 21 and  $\Delta(g_+) = 2\delta_a(p)$ ,  $G(Y^+) = \Theta$ .

Conversely, let  $c_b = g_+$ , and let  $c_j = \varepsilon_j s_a$  with  $\varepsilon_j = \pm 1$  and  $\psi_j = 0$  for  $j < b$ ; by Lemma 22, all cells of every  $\psi_j$  vanish. Put  $\beta_j = (M\zeta_j)_a$ , so that  $|\beta_j| \leq \omega_a = p^a$ . We show  $\varepsilon_j = (-1)^j$  by induction on  $j$  downwards. The cell of  $\psi_j$  at  $i = a$  has the arguments  $A = \varepsilon_j \delta_a(p)$  and  $B = \beta_j$ , and by (11) it is negative if  $AB > 0$  and  $|B| < P|A|$ . So it suffices to

check these two conditions for  $\varepsilon_j = (-1)^{j+1}$ . For  $j = b - 1$ ,  $\beta_{b-1} = (Mg_+)_a = (-1)^b \delta_a(p) - 2$  has the sign  $(-1)^b$ , and  $|\beta_{b-1}| \leq \delta_a(p) + 2 < P\delta_a(p)$ . For  $j \leq b - 2$ , if  $\varepsilon_{j+1} = (-1)^{j+1}$ , then  $\beta_j = (\beta_{j+1} + (-1)^{j+1} P\delta_a(p))/x$  has the sign  $(-1)^{j+1}$ , since  $|\beta_{j+1}| \leq p^a < P\delta_a(p)$ , and  $|\beta_j| \leq p^a < P\delta_a(p)$ . Hence  $\varepsilon_j = (-1)^j$  for all  $j < b$ , that is,  $Y = Y^+$ .  $\square$

## 11 Proof of Theorem 4

**Theorem 25.** *Let  $b \geq 2$  be even, let  $p, q$  be real numbers with  $p \geq 5$  and  $q \geq 3$ , or with  $p = 3$  and  $q \geq 5$ , and put  $\Theta = d_2(p)d_b(q) - 2\delta_2(p)\delta_b(q)$ . Every  $Y \in \{\pm 1\}^{3 \times (b+1)}$  with  $y_{2b} = -1$  satisfies  $G(Y) \leq \Theta$ , where  $G(Y)$  is the middle expression in (2) with  $Z = T_2(p)Y T_b(q)^\top$ , and equality holds exactly for  $Y = -s_2 s_b^\top$  and  $Y = s_2 s_b^\top - 2e_2 e_b^\top$ .*

*Proof of Theorem 4.* Let  $n = p^2 q^b$ . By (2),  $2\mathcal{E}(\text{ICG}_n(\mathcal{D})) = n + G(Y)$  for the sign matrix  $Y$  of a set  $\mathcal{D}$  of proper divisors of  $n$ , and  $y_{2b} = -1$ . The two matrices in Theorem 25 are the sign matrices of  $\{p^i q^j : i + j \text{ odd}\}$  and of  $\{p^i q^j : i + j \text{ even}\} \setminus \{n\}$ . These are nonempty sets of proper divisors of  $n$ : the first contains  $p$ , and it does not contain  $n$  because  $2 + b$  is even, and the second contains 1. Distinct odd primes satisfy the hypothesis of Theorem 25, so the maximal energy is  $\frac{1}{2}(n + \Theta)$ , attained exactly by these two sets. By Remark 9,  $d_2(p) = 5p^2 - 8p + 4$  and  $\delta_2(p) = p^2 - 2p + 2$ . The case  $n = p^b q^2$  follows by exchanging the names of the primes.  $\square$

We now prove Theorem 25, with the notation of Section 10 for  $a = 2$ . Here

$$M = T_2(p) = \begin{pmatrix} 0 & -pR & pR \\ -pR & R^2 & R \\ pR & R & 1 \end{pmatrix},$$

and we put  $g_1 = s_2 = (1, -1, 1)^\top$ ,  $g_2 = (1, 1, -1)^\top$ ,  $g_3 = (1, -1, -1)^\top = g_+$  and  $g_4 = (1, 1, 1)^\top$ . Using  $p = R + 1$ ,

$$Mg_1 = \begin{pmatrix} 2pR \\ -2R^2 \\ R^2 + 1 \end{pmatrix}, \quad Mg_2 = \begin{pmatrix} -2pR \\ -2R \\ R^2 + 2R - 1 \end{pmatrix}, \quad Mg_3 = \begin{pmatrix} 0 \\ -2pR \\ R^2 - 1 \end{pmatrix}, \quad Mg_4 = \begin{pmatrix} 0 \\ 0 \\ p^2 \end{pmatrix}.$$

Every  $c \in \{\pm 1\}^3$  equals  $\pm g_k$  for exactly one  $k$ . As  $d_2(p) = 5R^2 + 2R + 1$  and  $\delta_2(p) = R^2 + 1$ , we get  $\Delta(\pm g_k) = \Delta_k$  with

$$\Delta_1 = 0, \quad \Delta_2 = 2(R - 1)^2, \quad \Delta_3 = 2(R^2 + 1) = 2\delta_2(p), \quad \Delta_4 = 4R^2. \quad (14)$$

Moreover  $\omega = (2pR, 2pR, p^2)$ , and

$$p^2 < P(R^2 + 2R - 1), \quad p^2 < P(R^2 - 1), \quad (15)$$

because  $2(R^2 + 2R - 1) - p^2 = (R + 3)(R - 1) > 0$ ,  $2(R^2 - 1) - p^2 = (R + 1)(R - 3) > 0$  if  $R \geq 4$ , and  $P(R^2 - 1) \geq 12 > 9 = p^2$  if  $R = 2$ .

**Lemma 26.** *Let  $0 \leq \mu \leq 1$  and  $\zeta \in [-1, 1]^3$ , and put  $B = M\zeta$ . If  $c \neq \pm s_2$ , then  $\psi_\mu(c, \zeta) > 0$ . Moreover*

$$\psi_\mu(\pm g_3, \zeta) \geq 2P(1 + \mu)(R^2 + 1) - 2\mu|B_0|, \quad \psi_\mu(\pm g_4, \zeta) \geq 4PR^2(1 + \mu) - 2\mu(|B_0| + |B_1|). \quad (16)$$

*Proof.* We use Lemma 22,  $|B_i| \leq \omega_i$  and (15). For  $c = \pm g_2$  we have  $|Mc| = (2pR, 2R, R^2 + 2R - 1)$ , so the terms with  $i = 0, 2$  of the sum in Lemma 22 vanish, and  $\psi_\mu(c, \zeta) \geq 2P(1 + \mu)(R - 1)^2 - 4\mu R \max\{0, p - P\}$ . If  $P \geq p$ , this is positive. Otherwise  $R \geq 4$  by (8), and the bound is affine in  $\mu$ , positive at  $\mu = 0$ , and at  $\mu = 1$  equal to  $4[P(R^2 - R + 1) - R(R + 1)] \geq 4(R - 1)(R - 2) > 0$ . For  $c = \pm g_3$  we have  $|Mc| = (0, 2pR, R^2 - 1)$ , so the terms with  $i = 1, 2$  vanish, which gives the first bound in (16); with  $|B_0| \leq 2pR$  and  $P \geq 2$  its right side is at least  $4(1 + \mu)(R^2 + 1) - 4\mu pR = 4(R^2 + 1) - 4\mu(R - 1) > 0$ . For  $c = \pm g_4$  we have  $|Mc| = (0, 0, p^2)$ , which gives the second bound; its right side is at least  $8R^2(1 + \mu) - 8\mu pR = 8R(R - \mu) > 0$ .  $\square$

**Lemma 27.** Let  $R \geq 4$  and  $P \geq 2$ , or  $R = 2$  and  $P \geq 4$ , and put  $T = (R^2 + 1)(P^2 + 1)$ . Then:

- (i)  $(P - 1)^2(5R^2 + 2R + 1) > T$ ;
- (ii)  $P(P + 1)(R - 1)^2 + (P - 1)^2(3R^2 + 1) > T$ ;
- (iii)  $(R^2 + 1)(P - 1) + (P + 1)[2P(R^2 - R + 1) - 3R^2 - 1] > 0$ ;
- (iv)  $P(P + 1)(R - 1)^2 + 2(P - 1)^2R(R + 1) > T$ ;
- (v)  $(P + 1)[P(R - 1)^2 + 2P^2(R + 1)] > T$ ;
- (vi)  $(P + 1)[2PR(R - 3) + 2P^2(R + 1)] > T$  if  $R \geq 4$ ;
- (vii)  $P(P + 1)^2(R^2 + 1) > 2R(R + 1)(P^2 + 1)$ ;
- (viii)  $2R(PR - 1)(P + 1) > T$ .

*Proof.* The function  $f(P) = (P - 1)^2/(P^2 + 1) = 1 - 2P/(P^2 + 1)$  increases for  $P \geq 1$ , and  $f(2) = \frac{1}{5}$ ,  $f(4) = \frac{9}{17}$ . (i) We have  $5R^2 + 2R + 1 - 5(R^2 + 1) = 2(R - 2) \geq 0$  and  $5(P - 1)^2 - (P^2 + 1) = 2(2P - 1)(P - 2) \geq 0$ , so  $(P - 1)^2(5R^2 + 2R + 1) \geq 5(P - 1)^2(R^2 + 1) \geq T$ , and one of the two inequalities is strict. (ii) The left side minus  $T$  equals  $(R - 1)^2(P - 1) + (P - 1)^2(3R^2 + 1) - 2R(P^2 + 1)$ , and  $(P - 1)^2(3R^2 + 1) = f(P)(3R^2 + 1)(P^2 + 1) \geq 2R(P^2 + 1)$ : if  $R \geq 4$ , then  $f(P) \geq \frac{1}{5}$  and  $3R^2 + 1 - 10R = R(3R - 10) + 1 > 0$ ; if  $R = 2$ , then  $f(P) \geq \frac{9}{17}$  and  $\frac{9}{17} \cdot 13 > 4$ . (iii) If  $R \geq 4$ , then  $2P(R^2 - R + 1) - 3R^2 - 1 \geq 4(R^2 - R + 1) - 3R^2 - 1 = (R - 1)(R - 3) > 0$ ; if  $R = 2$ , it equals  $6P - 13 > 0$ . (iv) The left side minus  $T$  equals  $(R - 1)^2(P - 1) + 2R[(P - 1)^2(R + 1) - (P^2 + 1)]$ , and  $f(P)(R + 1) \geq 1$  both for  $R \geq 4$  and for  $R = 2$ . (v) The left side minus  $T$  equals  $(R - 1)^2(P - 1) + 2[R(P^3 - 1) + P^3 + P^2] > 0$ . (vi) As  $P(P + 1) \geq P^2 + 1$ ,  $R \geq 3$ ,  $P^2 \geq \frac{4}{5}(P^2 + 1)$  and  $P + 1 \geq 3$ , the left side is at least  $(P^2 + 1)[2R(R - 3) + \frac{24}{5}(R + 1)]$ , and  $2R(R - 3) + \frac{24}{5}(R + 1) - (R^2 + 1) = R^2 - \frac{6}{5}R + \frac{19}{5} > 0$ . (vii)  $P(P + 1)^2 = (P + 2)(P^2 + 1) - 2$ , so  $P(P + 1)^2 \geq \frac{18}{5}(P^2 + 1)$  for  $P \geq 2$ , and  $\frac{18}{5}(R^2 + 1) - 2R(R + 1) = \frac{1}{5}(8R^2 - 10R + 18) > 0$ . (viii) Twice the left side minus  $2T$  equals  $PR(PR - 4) + P^2(R^2 - 2) + (4PR^2 - 2R^2 - 4R - 2)$ , where  $PR \geq 8$ ,  $R \geq 2$  and  $4PR^2 - 2R^2 - 4R - 2 \geq 2(3R + 1)(R - 1) > 0$ .  $\square$

Each of these inequalities also follows from the substitutions  $R = 4 + s$ ,  $P = 2 + t$  and  $R = 2$ ,  $P = 4 + t$  with  $s, t \geq 0$ : in both cases the left side minus the right side is a polynomial in  $s$  and  $t$  with nonnegative coefficients and a positive constant term. This is how they are proved in the Lean formalisation (Section 13), where  $R$  and  $P$  are named  $P$  and  $Q$ , respectively,  $s$  and  $t$  are written as squares, and (iii)–(vi) and (viii) appear multiplied by 2.

**Lemma 28.** Let  $b$  be even,  $0 \leq J < b$ ,  $\mu = \mu_{J-1}$  and  $m = \mu_{b-2-J}$ . For every  $c \in \{\pm 1\}^3$  other than  $-s_2$ ,  $\psi_\mu(c, ms_2) > 2x\delta_2(p)\rho$ .

*Proof.* By (9),  $\mu, m \in [\mu_0, 1]$  and  $\rho \leq \mu_1 = (P^2 + 1)/x^2$ . Put  $T = (R^2 + 1)(P^2 + 1)$  and  $\psi = \psi_\mu(c, ms_2)$ . Then  $2x\delta_2(p)\rho \leq 2T/x$ , so it suffices to show  $x\psi > 2T$ ; we do so except in one subcase of (d). We use

$$x(P - \mu)m \geq (P - 1)^2, \quad (17)$$

which holds as  $P - \mu \geq P - 1 \geq 0$  and  $xm \geq x\mu_0 = P - 1$ . Let  $A = Mc$  and  $B = M(ms_2) = m(2pR, -2R^2, R^2 + 1)$ , and write  $h_i = h_\mu(A_i, B_i)$ ; the cells are computed with the values of  $h_\mu$  in Section 9 and (10), using  $0 \leq m \leq 1$ .

- (a)  $c = s_2$ . For each  $i$ ,  $A_i B_i \geq 0$  and  $|B_i| \leq |A_i|$ , so  $h_i = -2(P - \mu)m|A_i|$  and  $\psi = 2(P - \mu)m d_2(p)$ . By (17) and Lemma 27(i),  $x\psi \geq 2(P - 1)^2(5R^2 + 2R + 1) > 2T$ .

- (b)  $c = g_2$ , so  $A = (-2pR, -2R, R^2 + 2R - 1)$ . Here  $h_0 = 0$ , as  $A_0B_0 \leq 0$  and  $|B_0| \leq P|A_0|$ , and  $h_2 = -2(P - \mu)m(R^2 + 1)$ , as  $A_2B_2 \geq 0$  and  $|B_2| \leq |A_2|$ ; also  $A_1B_1 \geq 0$ . If  $mR \leq 1$ , then  $|B_1| \leq |A_1|$ ,  $h_1 = -4(P - \mu)mR^2$  and  $\psi = 2P(1 + \mu)(R - 1)^2 + 2(P - \mu)m(3R^2 + 1)$ , so  $x\psi > 2T$  by  $\mu \geq 0$ , (17) and Lemma 27(ii). If  $mR > 1$ , then  $h_1 = 4R(\mu mR - P)$  by (10), and expanding gives

$$x\psi - 2T = 2(R^2 + 1)(P - 1) + 2Px(R - 1)^2\mu + 2Px(R^2 + 1)m - 2x(3R^2 + 1)\mu m.$$

This is affine in  $\mu$  and in  $m$  separately, so its minimum over  $[0, 1]^2$  is attained at a corner. It is positive at the corners with  $\mu m = 0$ , and at  $\mu = m = 1$  it equals twice the left side of Lemma 27(iii).

- (c)  $c = -g_2$ , so  $A = (2pR, 2R, -(R^2 + 2R - 1))$ . Here  $h_0 = -4(P - \mu)mpR$  and  $h_2 = 0$ , and  $A_1B_1 \leq 0$ . If  $mR \leq P$ , then  $|B_1| \leq P|A_1|$ ,  $h_1 = 0$  and  $\psi = 2P(1 + \mu)(R - 1)^2 + 4(P - \mu)mpR$ , so  $x\psi > 2T$  by (17) and Lemma 27(iv). If  $mR > P$ , then  $R \geq 4$ , since  $mR \leq 2 < P$  when  $R = 2$ ; now  $h_1 = 4\mu R(mR - P)$  by (10), and

$$\psi = 2P(1 + \mu)(R - 1)^2 + 4\mu PR + 4mR(Pp - \mu(2R + 1)).$$

As  $Pp - \mu(2R + 1) \geq 2(R + 1) - (2R + 1) > 0$  and  $mR > P$ , we get  $\psi > 2P(R - 1)^2 + 4P^2(R + 1) + 2\mu P(R^2 - 4R - 1)$ . The right side is affine in  $\mu$ , and  $x$  times it exceeds  $2T$  at  $\mu = 0$  by Lemma 27(v) and at  $\mu = 1$  by Lemma 27(vi), since  $(R - 1)^2 + R^2 - 4R - 1 = 2R(R - 3)$ .

- (d)  $c = \pm g_3$ . By (16) with  $|B_0| = 2mpR$ ,  $\psi \geq 2P(1 + \mu)(R^2 + 1) - 4\mu mpR$ . If  $J \leq b - 2$ , then  $m \leq \mu_1$  by (9), so  $2mpR \leq 2pR(P^2 + 1)/x^2 < P(R^2 + 1)$  by Lemma 27(vii); hence  $\psi \geq 2P(R^2 + 1)$  and  $x\psi \geq 2P(P + 1)(R^2 + 1) > 2T$ . If  $J = b - 1$ , then  $m = \mu_{-1} = 1$ ,  $\mu = \mu_{b-2} > 0$  and  $x\rho = P - \mu$ , and directly

$$\psi - 2x\delta_2(p)\rho \geq 2\mu[(R^2 + 1)x - 2pR] \geq 2\mu(R^2 - 2R + 3) > 0.$$

- (e)  $c = \pm g_4$ . By (16) with  $|B_0| + |B_1| = 2mR(2R + 1)$  and  $m \leq 1$ ,  $\psi \geq 4PR^2(1 + \mu) - 4\mu R(2R + 1) = 4R(PR - 1) + 4R[\mu R(P - 2) + 1 - \mu] \geq 4R(PR - 1)$ , so  $x\psi > 2T$  by Lemma 27(viii).

These are all the columns  $c \neq -s_2$ . □

**Lemma 29.** *Let  $b$  be even. If  $c_b = g_2$ , then  $\sum_{j < b} \psi_j > 4Rx\rho$ .*

*Proof.* All  $\psi_j$  are nonnegative by Lemmas 22 and 26. Here  $\zeta_{b-1} = g_2$ , so  $B = M\zeta_{b-1} = (-2pR, -2R, R^2 + 2R - 1)$ , and  $x\rho = P - \mu$  with  $\mu = \mu_{b-2}$ . Let  $c = c_{b-1}$ ,  $A = Mc$  and  $h_i = h_\mu(A_i, B_i)$ .

Suppose first that  $c \neq -g_2$  or  $R \geq 4$ . We show that  $\psi_{b-1} > 4R(P - \mu)$ .

- $c = \pm g_4$ : by (16),  $\psi_{b-1} - 4R(P - \mu) \geq 4PR^2(1 + \mu) - 4\mu R(R + 2) - 4R(P - \mu) = 4PR(R - 1) + 4\mu R(PR - R - 1) > 0$ .
- $c = \pm g_3$ : by (16),  $\psi_{b-1} - 4R(P - \mu) \geq 2P(1 + \mu)(R^2 + 1) - 4\mu pR - 4R(P - \mu) = 2P(R - 1)^2 + 2\mu[P(R^2 + 1) - 2R^2] > 0$ .
- $c = g_2$  or  $c = -s_2$ : then  $A_0 = B_0 < 0$ , so  $h_0 = -4(P - \mu)pR$ , while  $h_1, h_2 \leq 0$ , for  $g_2$  because  $A_i = B_i$  and for  $-s_2$  by Lemma 22. Hence  $\psi_{b-1} \geq 4(P - \mu)pR > 4R(P - \mu)$ .
- $c = s_2$ : here  $A = (2pR, -2R^2, R^2 + 1)$ , so  $h_0 = 0$ ,  $h_1 = -4(P - \mu)R$  and  $h_2 = 2\mu(R^2 + 2R - 1) - 2P(R^2 + 1)$  by (10). Hence  $\psi_{b-1} - 4R(P - \mu) = 2P(R^2 + 1) - 2\mu(R^2 + 2R - 1) \geq 2(R^2 - 2R + 3) > 0$ .

- $c = -g_2$  and  $R \geq 4$ : by Lemma 23,  $\psi_{b-1} = 2P(1 + \mu)(R - 1)^2$ , and  $\psi_{b-1} - 4R(P - \mu) \geq 2P(R^2 - 4R + 1) > 0$ .

Now let  $c = -g_2$  and  $R = 2$ , so that  $p = 3$  and  $P \geq 4$ . By Lemma 23,  $\psi_{b-1} = 2P(1 + \mu_{b-2})$  and  $\zeta_{b-2} = -\mu_0 g_2$ , so  $B' = M\zeta_{b-2} = m(12, 4, -7)$  with  $m = \mu_0 < 1$ . We show that  $\psi_{b-2} \geq 2P(1 + \mu')$ , where  $\mu' = \mu_{b-3}$ , for every column  $c' = c_{b-2}$ . Here  $Mg_1 = (12, -8, 5)$ ,  $Mg_2 = (-12, -4, 7)$ ,  $Mg_3 = (0, -12, 3)$  and  $Mg_4 = (0, 0, 9)$ , and

$$22(P - \mu')m \geq 2P(1 + \mu'), \quad (18)$$

because  $x \cdot 22(P - \mu')m = 22(P - \mu')(P - 1) \geq 22(P - 1)^2 \geq 4P(P + 1) \geq 2P(1 + \mu')x$ ; indeed  $22(P - 1)^2 - 4P(P + 1) = 18P^2 - 48P + 22 > 0$  for  $P \geq 4$ . Let  $h'_i$  be the cells of  $\psi_{b-2}$ .

- $c' = \pm g_4$ : by (16),  $\psi_{b-2} \geq 16P(1 + \mu') - 32\mu'm \geq 2P(1 + \mu') + 14P(1 + \mu') - 32\mu' > 2P(1 + \mu')$ .
- $c' = \pm g_3$ : by (16),  $\psi_{b-2} \geq 10P(1 + \mu') - 24\mu'm > 2P(1 + \mu')$ .
- $c' = \pm g_2$ :  $|B'_i| \leq |(Mc')_i|$  for all  $i$ , so all cells are  $\leq 0$  and  $\psi_{b-2} \geq P(1 + \mu')\Delta_2 = 2P(1 + \mu')$ .
- $c' = s_2$ :  $h'_0 = -24(P - \mu')m$  and  $h'_1, h'_2 \leq 0$  by Lemma 22, so  $\psi_{b-2} \geq 24(P - \mu')m$ , and (18) applies.
- $c' = -s_2$ : then  $Mc' = (-12, 8, -5)$ ,  $h'_0 \leq 0$  and  $h'_1 = -8(P - \mu')m$ . If  $7m \leq 5$ , then  $h'_2 = -14(P - \mu')m$  and  $\psi_{b-2} \geq 22(P - \mu')m$ . If  $7m > 5$ , then  $h'_2 = 14\mu'm - 10P$  by (10), and  $\psi_{b-2} \geq 8(P - \mu')m - 14\mu'm + 10P \geq 8Pm - 22m + 10P > 4P \geq 2P(1 + \mu')$ .

Finally put  $u = \mu_{b-2}$  and  $v = \mu_{b-3}$ , so that  $xu = P - v$  and  $x\rho = P - u$ . By (4),  $v > \mu_\infty = P/(P + 2)$ , since  $b - 3$  is odd or  $v = \mu_{-1} = 1$ . Using  $xu = P - v$ , one finds

$$x[2P(1 + u) + 2P(1 + v) - 8(P - u)] = 2(P - 2)[(P + 2)v - P] > 0.$$

Hence  $\psi_{b-1} + \psi_{b-2} \geq 2P(1 + u) + 2P(1 + v) > 8x\rho = 4Rx\rho$ .  $\square$

*Proof of Theorem 25.* Let  $Y \in \{\pm 1\}^{3 \times (b+1)}$  with  $y_{2b} = -1$ . By Lemmas 22 and 26, all  $\psi_j$  are nonnegative, so Lemma 21 gives

$$x^{1-b}(\Theta - G(Y)) \geq \sum_{j < b} \psi_j + x\rho(\Delta(c_b) - 2\delta_2(p)). \quad (19)$$

If  $Y = Y^-$ , then  $G(Y) = \Theta$ . Otherwise let  $J$  be the largest index with  $c_J \neq (-1)^{J+1}s_2$ .

If  $J < b$ , then  $c_b = -s_2$  as  $b$  is even, so  $\Delta(c_b) = 0$ . By Lemma 23,  $\zeta_J = \varepsilon\mu_{b-2-J}s_2$  with  $\varepsilon = (-1)^J$ , so  $\psi_J = \psi_{\mu_{J-1}}(\varepsilon c_J, \mu_{b-2-J}s_2)$  with  $\varepsilon c_J \neq -s_2$ . By Lemma 28,  $\psi_J > 2x\delta_2(p)\rho$ , and (19) gives  $G(Y) < \Theta$ .

If  $J = b$ , then  $c_b \neq -s_2$  and  $y_{2b} = -1$ , so  $c_b$  is  $g_2, g_3$  or  $-g_4$ . If  $c_b = -g_4$ , then  $\Delta(c_b) - 2\delta_2(p) = 2R^2 - 2 > 0$  by (14), and  $G(Y) < \Theta$ . If  $c_b = g_2$ , then  $\Delta(c_b) - 2\delta_2(p) = -4R$ , and  $G(Y) < \Theta$  by Lemma 29. If  $c_b = g_3 = g_+$ , then Lemma 21 gives  $x^{1-b}(\Theta - G(Y)) = \sum_j \psi_j + 2x((M\zeta_{-1})_2)^- \geq 0$ . If equality holds, all  $\psi_j$  vanish, so every  $c_j$  with  $j < b$  is  $\pm s_2$  by Lemma 26, and  $Y = Y^+$  by Lemma 24. Conversely  $G(Y^+) = \Theta$  by Lemma 24.  $\square$

*Remark 30.* The argument has little room in two families. For  $b = 2$ ,  $q = 3$  and  $Y = s_2(1, 1, -1)^\top$ , which is case (a) of Lemma 28 with  $J = 0$ , we have  $\Theta - G(Y) = 4(p - 3)$  (Remark 17). For  $p = 3$  and  $Y = (1, 1, -1)^\top s_b^\top$ , whose columns are  $\pm g_2$ , we have  $\Theta - G(Y) = 2(d_b(q) - 5\delta_b(q))$ , which is  $4(q - 3)$  for  $b = 2$ . Both differences are small compared with  $\Theta$  when  $p$ , respectively  $q$ , is large.

## 12 Proof of Theorem 5

For  $\min\{a, b\} \leq 2$ , Theorem 5 is Theorem 3 or Theorem 4, after exchanging the names of the primes if necessary. Let  $3 \leq \min\{a, b\} \leq 8$ ; exchanging the names of the primes if necessary, we may assume that the exponent  $a$  of  $p$  satisfies  $3 \leq a \leq 8$ . As in the proof of Theorem 4, it suffices to show, for all real  $p, q$  satisfying (8) and all  $b \geq 1$  with  $b \equiv a \pmod{2}$ , that every  $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$  with  $y_{ab} = -1$  satisfies  $G(Y) \leq \Theta$ , with equality exactly for  $Y^-$  and  $Y^+$ . We use the notation of Section 10. The proof of Theorem 25 used  $a = 2$  only in Lemmas 26, 28 and 29 and in the list (14) of the possible last columns. Proposition 31 below replaces them by finitely many inequalities that do not depend on  $b$ ; for  $3 \leq a \leq 8$  these were verified by computer.

Let  $\mathcal{R}_1$  and  $\mathcal{R}_2$  be the two sets of pairs  $(p, q)$  in (8), and put  $P_{\min} = 2$  on  $\mathcal{R}_1$  and  $P_{\min} = 4$  on  $\mathcal{R}_2$ . For real  $y$  let  $\eta_0(y) = y$  and  $\eta_{n+1}(y) = (P - \eta_n(y))/x$ , so that  $\mu_{j+n} = \eta_n(\mu_j)$ . Put  $\hat{D}_1 = d_1(x)/x = (3x-4)/x$  and  $\hat{D}_2 = d_2(x)/x^2 = (5x^2-8x+4)/x^2$ . For  $a$  even, let  $\mathcal{F}_a$  be the set of the triples  $(\mu, m, r)$  of the forms  $(1, m, (P-m)/x)$  with  $\mu_0 \leq m \leq \mu_\infty$ ;  $(\mu, 1, (P-\mu)/x)$  with  $\mu_0 \leq \mu \leq \mu_\infty$ ;  $(\mu, m, \mu)$  with  $\mu_\infty \leq \mu \leq \mu_1$  and  $\mu_0 \leq m \leq \mu_\infty$ ; and  $(\mu, m, m)$  with  $\mu_0 \leq \mu \leq \mu_\infty$  and  $\mu_\infty \leq m \leq \mu_1$ . For  $a$  odd, let  $\mathcal{F}_a$  consist of  $(1, 1, \mu_0)$ ; the triples  $(1, m, (P-m)/x)$  and  $(\mu, 1, (P-\mu)/x)$  with  $\mu_\infty \leq m, \mu \leq \mu_1$ ; and the triples  $(\mu, m, \mu_\infty)$  with  $\mu, m$  both in  $[\mu_0, \mu_\infty]$  or both in  $[\mu_\infty, \mu_1]$ .

**Proposition 31.** *Let  $a \geq 2$ , and suppose that the following conditions hold at every point of  $\mathcal{R}_1$ , or at every point of  $\mathcal{R}_2$ .*

- (S<sub>a</sub>)  $P_{\min} \Delta(c) > \sum_{i=0}^a \max\{0, \omega_i - P_{\min}|(Mc)_i|\}$  for every  $c \in \{\pm 1\}^{a+1}$  other than  $\pm s_a$ .
- (F<sub>a</sub>)  $\psi_\mu(c, ms_a) > 2x\delta_a(p)r$  for every  $c \in \{\pm 1\}^{a+1}$  other than  $-s_a$  and every  $(\mu, m, r) \in \mathcal{F}_a$ .
- (C<sub>a</sub>) Every  $g \in \{\pm 1\}^{a+1}$  with  $g_a = -1$ , other than  $-(-1)^a s_a$  and  $g_+$ , satisfies  $\Delta(g) > 2\delta_a(p)$  or the following three conditions, where  $\gamma = 2\delta_a(p) - \Delta(g)$ .
  - (C1) For  $0 \leq k \leq 4$ , every  $y \in I_k$  and every  $c \in \{\pm 1\}^{a+1}$  other than  $(-1)^{k+1}g$ ,

$$\sum_{n=1}^k P(1 + \eta_n(y)) \Delta(g) + \psi_y(c, (-1)^k \mu_{k-1}g) > x \eta_{k+1}(y) \gamma,$$

where  $I_k = [\mu_0, \mu_\infty]$  if  $k \equiv a \pmod{2}$ , and  $I_k = [\mu_\infty, \mu_1] \cup \{1\}$  otherwise.

- (C2)  $\sum_{n=0}^4 P(1 + \eta_n(y)) \Delta(g) > x \eta_5(y) \gamma$  for every  $y \in [\mu_0, \mu_\infty]$  if  $a$  is even, and for every  $y \in [\mu_\infty, \mu_1]$  if  $a$  is odd.
- (C3)  $\Delta(g)(\hat{D}_1/\mu_1 + 2) > 2\delta_a(p)$  if  $a$  is even;  $\Delta(g)\hat{D}_1/\mu_0 > 2\delta_a(p)$  and  $\Delta(g)(\hat{D}_2/\mu_\infty + 2) > 2\delta_a(p)$  if  $a$  is odd.

Then at every point of that region and for every  $b \geq 1$  with  $b \equiv a \pmod{2}$ , every  $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$  with  $y_{ab} = -1$  satisfies  $G(Y) \leq \Theta$ , with equality exactly for  $Y^-$  and  $Y^+$ .

*Proof.* First,  $\psi_\mu(c, \zeta) > 0$  for all  $c \neq \pm s_a$ ,  $0 \leq \mu \leq 1$  and  $\zeta \in [-1, 1]^{a+1}$ . Indeed, by Lemma 22 and  $|(M\zeta)_i| \leq \omega_i$ ,  $\psi_\mu(c, \zeta)$  is at least  $P(1 + \mu)\Delta(c) - 2\mu \sum_i \max\{0, \omega_i - P|(Mc)_i|\}$ , which is affine in  $\mu$ ; at  $\mu = 0$  it equals  $P\Delta(c) > 0$ , and at  $\mu = 1$  it is at least twice the difference of the two sides of (S<sub>a</sub>), since  $P \geq P_{\min}$ . So, by Lemma 22, all  $\psi_j$  are nonnegative,  $\psi_j = 0$  only if  $c_j = \pm s_a$ , and Lemma 21 gives

$$x^{1-b}(\Theta - G(Y)) \geq \sum_{j < b} \psi_j + x\rho(\Delta(c_b) - 2\delta_a(p)). \quad (20)$$

If  $Y = Y^-$ , then  $G(Y) = \Theta$ . Otherwise let  $J$  be the largest index with  $c_J \neq (-1)^{J+1} s_a$ .

*Case  $J < b$ .* Then  $\Delta(c_b) = 0$ , and as in the proof of Theorem 25,  $\psi_J = \psi_\mu(\varepsilon c_J, m s_a)$  with  $\mu = \mu_{J-1}$ ,  $m = \mu_{b-2-J}$ ,  $\varepsilon = (-1)^J$  and  $\varepsilon c_J \neq -s_a$ . We claim that  $(\mu, m, r) \in \mathcal{F}_a$  for some  $r \geq \rho$ ; then  $\psi_J > 2x\delta_a(p)\rho$  by (F<sub>a</sub>), and  $G(Y) < \Theta$  by (20). Let  $a$ , and hence  $b$ , be even. If  $J = 0$ , then  $\mu = \mu_{-1} = 1$ ,  $m = \mu_{b-2} \in [\mu_0, \mu_\infty]$  by (4), and  $\rho = (P - m)/x$ . If  $J = b - 1$ , then  $m = 1$ ,  $\mu = \mu_{b-2}$  and  $\rho = (P - \mu)/x$ . If  $1 \leq J \leq b - 2$ , the indices  $J - 1$  and  $b - 2 - J$  are nonnegative with odd sum; by (4), the value with the even index lies in  $[\mu_0, \mu_\infty]$  and the value with the odd index lies in  $[\mu_\infty, \mu_1]$  and exceeds  $\rho = \mu_{b-1}$ , since the  $\mu_j$  with odd  $j$  decrease. Now let  $a$  be odd. If  $b = 1$ , then  $J = 0$  and  $(\mu, m, \rho) = (1, 1, \mu_0)$ . If  $b \geq 3$ , the cases  $J = 0$  and  $J = b - 1$  are as before, now with  $\mu_{b-2} \in [\mu_\infty, \mu_1]$ ; and if  $1 \leq J \leq b - 2$ , the indices  $J - 1$  and  $b - 2 - J$  have the same parity, while  $\rho = \mu_{b-1} < \mu_\infty$  as  $b - 1$  is even.

*Case  $J = b$ .* Then  $g = c_b$  satisfies  $g_a = -1$  and  $g \neq -(-1)^a s_a$ . If  $g = g_+$ , then Lemmas 21 and 24 give  $x^{1-b}(\Theta - G(Y)) = \sum_j \psi_j + 2x((M\zeta_{-1})_a)^- \geq 0$ ; equality forces all  $\psi_j$  to vanish, so that  $c_j = \pm s_a$  for  $j < b$  and  $Y = Y^+$  by Lemma 24, while  $G(Y^+) = \Theta$ . If  $\Delta(g) > 2\delta_a(p)$ , then  $G(Y) < \Theta$  by (20). Otherwise (C1)–(C3) hold, and by (20) it suffices to show  $\sum_j \psi_j > x\rho\gamma$ . Let  $k$  be the largest integer with  $k \leq b$  and  $c_{b-i} = (-1)^i g$  for  $1 \leq i \leq k$ . By Lemma 23,  $\psi_{b-i} = P(1 + \mu_{b-i-1})\Delta(g)$  for  $1 \leq i \leq k$ .

- If  $k < b$  and  $k \leq 4$ , put  $y = \mu_{b-2-k}$ , so that  $\mu_{b-i-1} = \eta_{k+1-i}(y)$  and  $\rho = \eta_{k+1}(y)$ . Then  $y \in I_k$ : if  $b-2-k \geq 0$ , by (4), as  $b-2-k \equiv a-k \pmod{2}$ ; and if  $b = k+1$ , then  $y = \mu_{-1} = 1$  and  $k \not\equiv a \pmod{2}$ . The column  $j = b-1-k$  satisfies  $c_j \neq (-1)^{k+1}g$  and, by Lemma 23,  $\zeta_j = (-1)^k \mu_{k-1}g$ ; so  $\psi_j = \psi_y(c_j, (-1)^k \mu_{k-1}g)$ , and (C1) gives  $\sum_{i=1}^k \psi_{b-i} + \psi_j > x\rho\gamma$ .
- If  $5 \leq k < b$ , then  $b \geq 6$ , and  $y = \mu_{b-6}$  lies in  $[\mu_0, \mu_\infty]$  if  $a$  is even and in  $[\mu_\infty, \mu_1]$  if  $a$  is odd. The columns  $b-1, \dots, b-5$  give  $\sum_{i=1}^5 \psi_{b-i} = \sum_{n=0}^4 P(1 + \eta_n(y))\Delta(g)$ , and  $\rho = \eta_5(y)$ , so (C2) applies.
- If  $k = b$ , then  $\sum_j \psi_j = x\Delta(g)(\hat{D} - \rho)$ , and the claim reads  $\Delta(g)\hat{D}/\rho > 2\delta_a(p)$ . By Remark 9,  $\hat{D} = \hat{D}_1 + 2\sum_{i=1}^{b-1} \mu_i$ . If  $b$  is even, then  $\rho = \mu_{b-1} \leq \mu_1$ , so  $\hat{D}/\rho \geq \hat{D}_1/\mu_1 + 2$ ; if  $b = 1$ , then  $\hat{D}/\rho = \hat{D}_1/\mu_0$ ; and if  $b \geq 3$  is odd, then  $\hat{D} \geq \hat{D}_2 + 2\mu_{b-1}$  and  $\rho < \mu_\infty$ , so  $\hat{D}/\rho \geq \hat{D}_2/\mu_\infty + 2$ . Since  $\Delta(g) > 0$ , (C3) applies.  $\square$

**The certificates.** For  $3 \leq a \leq 8$  we verified (S<sub>a</sub>), (F<sub>a</sub>) and (C<sub>a</sub>) on both regions by exact computer calculation. The entries of  $M$ ,  $Mc$ ,  $\alpha$  and  $Mg$  are polynomials in  $p$ , while  $\mu_0, \mu_\infty, \mu_1$  and the maps  $\eta_n$  are rational in  $q$ . Consider (F<sub>a</sub>) for a fixed  $c$  and one of the families of triples in  $\mathcal{F}_a$ . The programs first certify the sign of every entry of  $Mc$  and of  $\alpha$  on the region, and they stop if a sign is not constant. By the values of  $h_\mu$  in Section 9 and (10), minus the cell  $h_\mu((Mc)_i, m\alpha_i)$  then equals  $-2\mu m|\alpha_i|$  if  $(Mc)_i = 0$ ; 0 if  $\alpha_i = 0 \neq (Mc)_i$ ;  $2\min\{(P - \mu)m|\alpha_i|, P|(Mc)_i| - \mu m|\alpha_i|\}$  if  $(Mc)_i\alpha_i > 0$ ; and  $-2\mu \max\{0, m|\alpha_i| - P|(Mc)_i|\}$  if  $(Mc)_i\alpha_i < 0$ . So  $\psi_\mu(c, m s_a) - 2x\delta_a(p)r$  is a minimum of finitely many functions, each affine in  $\mu$  for fixed  $m$  and affine in  $m$  for fixed  $\mu$ ; it is therefore concave in each variable, and its minimum over each family is attained at a corner, where  $\mu$  and  $m$  take values among  $\mu_0, \mu_\infty, \mu_1$  and 1. At a corner it is positive exactly when each of the finitely many branches, obtained by choosing one term in every minimum, is positive. Each branch is a rational function of  $p$  and  $q$  whose denominator is a product of powers of  $x$  and  $x + 1$ . It is certified positive by substituting  $p = 5 + s$ ,  $q = 3 + t$  on  $\mathcal{R}_1$ , or  $p = 3$ ,  $q = 5 + t$  on  $\mathcal{R}_2$ , and checking that the numerator is a polynomial in  $s, t$  with nonnegative coefficients and a positive constant term. Condition (C1) is treated in the same way, with the parameter  $y$ , on which the sum over the chain and  $\eta_{k+1}(y)$  depend affinely. Conditions (S<sub>a</sub>), (C2) and (C3), and the sign of  $\Delta(g) - 2\delta_a(p)$  for each last column  $g$ , are certified directly by the same coefficient test; as (C2) is affine in  $y$ , it is checked at the endpoints of its interval.

The last columns that need (C<sub>a</sub>) are  $(-1, 1, 1, -1)$  and  $(-1, -1, 1, -1)$  for  $a = 3$ , together with  $(1, 1, 1, -1)$  on  $\mathcal{R}_2$ , and  $(1, 1, -1, 1, -1)$ ,  $(1, -1, 1, 1, -1)$  and  $(1, -1, -1, 1, -1)$  for  $a = 4$ ,

together with  $(-1, -1, 1, 1, -1)$  on  $\mathcal{R}_2$ ; every other last column  $g$  with  $g_a = -1$ , other than  $-(-1)^a s_a$  and  $g_+$ , has  $\Delta(g) > 2\delta_a(p)$  on the region. The program `generic_prover.py`, with the parallel driver `generic_prover_fast.py`, generates the branches for  $(F_a)$  and  $(C1)$  and certifies them with SymPy: for  $a = 3$  there are 4 056 for  $(F_3)$  and 10 140 for  $(C1)$ , 14 196 in all, and for  $a = 4$  there are 15 648 and 54 768, 70 416 in all. It checks  $(S_a)$ ,  $(C2)$  and  $(C3)$  directly. All certificates succeed. The program `generic_fast2.py` repeats the computation with its own exact integer arithmetic for polynomials over the denominators  $x^k(x+1)^l$ , and obtains the same lists, again without failures. Independently, a referee agent wrote `c3_certify.py` (with `r2poly.py`). It does not use the formula for the cells above, but expands every cell from the definition of  $h_\mu$ , splitting each absolute value whose sign is not constant on the region into its two branches, and it evaluates every certified expression at a random rational point as a check. It certifies the complete conditions for  $a = 3$  and  $a = 4$  without failures. As negative controls, it rejects them when the right-hand side of  $(F_a)$  is multiplied by  $\frac{5}{4}$ , and on regions enlarged to contain  $p = q = 3$ .

The SymPy program `generic_prover.py` was run only for  $a \leq 4$ . For  $a = 5, 6, 7$  and  $8$  the lists were generated and certified by `generic_fast2.py`, with 69 368, 259 104, 1 129 336 and 4 182 048 branches for  $(F_a)$  and 381 524, 2 202 384, 11 858 028 and 54 366 624 for  $(C1)$ , and independently by `c3_certify.py`, all without failures. The last columns that need  $(C_a)$  number 4, 5, 6 and 7 on  $\mathcal{R}_1$  and 7, 12, 15 and 19 on  $\mathcal{R}_2$ ; they are listed in the logs of `c3_certify.py` and `rv3_certify.py` (directories `review2/logs` and `review3/logs` of [1]), and every other last column  $g$  with  $g_a = -1$ , other than  $-(-1)^a s_a$  and  $g_+$ , satisfies  $\Delta(g) > 2\delta_a(p)$  on the region. A third program, `rv3_certify.py`, was written by another referee agent. It builds  $T_a(p)$  from the Ramanujan-sum definition, enumerates the conditions of Proposition 31 from its statement, evaluates each cell by the case analysis in the proof of Lemma 10, and checks a sample of the certified expressions against a direct evaluation from the definitions at random exact points. It certifies  $(S_a)$ ,  $(F_a)$  and  $(C_a)$  for  $2 \leq a \leq 8$  on both regions without failures. For  $5 \leq a \leq 8$  this referee also checked, item by item, that `generic_fast2.py` certifies every instance of  $(F_a)$  and  $(C1)$  that Proposition 31 requires, and that the three programs select the same last columns. As negative controls for  $5 \leq a \leq 8$ , the certificates of `rv3_certify.py` fail when  $2\delta_a(p)$  is replaced by  $2.002\delta_a(p)$ , exactly at the conditions that correspond to  $Y^+$ ; they also fail for  $b$  of the other parity, and on regions containing  $p = q = 3$  or, for  $a \leq 7$ ,  $q = 2$ .

These programs and their logs are in the directory `certificates/icg-equal-parity/theorems-b-c` of [1], the referee agents' programs in its subdirectories `review2` and `review3`. The proof of Theorem 5 for  $3 \leq \min\{a, b\} \leq 8$  rests on these computations; it is not formalised.

**Checks.** For  $a = 3$  and  $a = 4$  the conclusion of Proposition 31 was also checked directly from  $Z = MYN^T$ , by complete enumeration of the sign matrices of the shapes  $(3, 7)$  at  $(p, q) = (5, 3)$ ,  $(3, 5)$  and  $(7, 3)$ , with  $2^{31}$  matrices each, and  $(4, 6)$  at  $(5, 3)$  and  $(3, 5)$ , with  $2^{34}$  matrices each (`t7_brute_numpy.py`). For larger  $b$  we used an exact branch-and-bound search (`bnb.py`). It bounds  $G$  with Lemmas 21 and 22, checks  $(S_a)$  exactly at the given parameters, and recomputes every sign matrix it reports from the matrix model. It agreed with complete enumeration in all 41 comparisons made (`t5_bnb_validate.py`). In 786 runs at integer and rational points of both regions (`t6_bnb_runs.py`, `t9_bnb_random.py`), for the shapes  $(3, b)$  with odd  $b \leq 21$  or  $b \in \{25, 31, 41\}$  and  $(4, b)$  with even  $b \leq 16$  or  $b \in \{20, 30, 40\}$ , it found the maximum  $\Theta$ , attained exactly at  $Y^-$  and  $Y^+$ .

For  $5 \leq a \leq 8$  the conclusion of Proposition 31 was checked by an exact branch-and-bound search (`rv3_bnb.py`, written independently of `bnb.py`) in 208 runs on the shapes  $(5, 5)$ ,  $(5, 7)$ ,  $(7, 5)$ ,  $(5, 9)$ ,  $(5, 11)$ ,  $(6, 6)$ ,  $(6, 8)$ ,  $(8, 6)$ ,  $(6, 10)$ ,  $(7, 7)$ ,  $(7, 9)$ ,  $(8, 8)$ ,  $(8, 10)$ ,  $(5, 25)$ ,  $(6, 20)$ ,  $(7, 15)$  and  $(8, 16)$ , at prime and rational points of both regions. In every run the maximum is  $\Theta$ , attained exactly at  $Y^-$  and  $Y^+$ . The search was validated against complete enumeration (`rv3_brute.py`) for the shapes  $(5, 1)$ ,  $(5, 3)$ ,  $(6, 2)$ ,  $(7, 1)$ ,  $(7, 3)$  and  $(8, 2)$ , with up to  $2^{31}$

matrices. The case analysis in the proof of Proposition 31 was also replayed in exact arithmetic (`rv3_replay.py`) on 1 600 random sign matrices with  $5 \leq a \leq 8$  and  $b \leq 13$ , at rational points of both regions, and every step held.

## 13 Verification

The proofs above are complete, except that the proofs of Proposition 15 and of the case  $3 \leq \min\{a, b\} \leq 8$  of Theorem 5 rest on computer calculations, described in the paragraph on certificates below and in Section 12. The programs and logs for Theorems 4 and 5, including those of the three referee agents, are in the directory `certificates/icg-equal-parity/theorems-b-c` of [1]. The following computations are independent cross-checks.

- An exhaustive search over all divisor sets, in exact integer arithmetic, confirms Theorem 2, including the value of the maximum, in the following cases:
  - exponent pairs  $(4, 1)$ ,  $(6, 1)$ ,  $(8, 1)$ ,  $(2, 5)$ ,  $(4, 3)$ , for the primes  $5 \leq p \leq 31$ ;
  - $(10, 1)$ ,  $(12, 1)$ ,  $(2, 7)$ ,  $(6, 3)$ , for  $p \in \{5, 7, 11, 13, 17\}$ ;
  - $(14, 1)$ ,  $(4, 5)$ ,  $(2, 9)$ , for  $p \in \{5, 7, 11\}$  (the largest case has  $2^{29}$  divisor sets).
- Theorem 12 was checked in exact arithmetic over all sign matrices of every shape with  $(a + 1)(b + 1) \leq 25$ , for  $(p, q) = (5, 3)$ ,  $(7, 3)$  and  $(5, 4)$ .
- Each lemma was tested separately in exact arithmetic: by adversarial searches for small  $b$ , and by structured random tests of Lemma 11 up to  $b = 20$ .
- Proposition 6 was compared, in floating-point arithmetic, with spectra of the actual circulant matrices. The fast Fourier transform was used for all divisor sets of the orders  $n = 75, 147, 363, 675, 1323, 1875$  and  $6075$ . Dense diagonalisation was used for all divisor sets when  $n \leq 675$  and for samples of them when  $n = 1323, 1875$  and  $6075$ .
- Theorem 3 was confirmed by exhaustive searches over all divisor sets: in exact arithmetic, with the eigenvalues computed from Ramanujan sums and independently of Proposition 6, for 35 orders  $pq^m$  with  $m \in \{1, 3, 5, 7\}$  up to  $n = 234375$ ; and with floating-point spectra from the fast Fourier transform for ten orders up to  $n = 9375$ .
- The inequality  $\Phi \leq \Theta$  of Section 9, with its equality cases, was checked in exact arithmetic over all pairs of sign vectors for  $m \leq 11$  at several parameter pairs  $(x, R)$ . Each intermediate claim of its proof was checked separately for  $m \leq 7$ , including non-integer parameters.
- An exhaustive search over all divisor sets confirms Conjecture 14 for ten ordered pairs of odd primes and each of the exponent pairs  $(1, 1)$ ,  $(2, 2)$ ,  $(1, 3)$ ,  $(3, 1)$ ,  $(3, 3)$ ,  $(2, 4)$ ,  $(4, 2)$ ,  $(4, 4)$ ,  $(1, 5)$  and  $(3, 5)$ . These cases are now covered by Theorems 3, 4 and 5 and by Proposition 15.
- Theorem 25 was confirmed by exhaustive maximisation of  $G$  over all sign matrices, in exact arithmetic, in 69 runs with  $b \in \{2, 4, 6, 8\}$  at prime and at non-integer parameters. The main steps of its proof in Sections 10 and 11 were replayed in exact arithmetic on all sign matrices with  $b \in \{2, 4\}$  at ten parameter pairs, including non-integer ones, and on random sign matrices with  $b \in \{6, 8\}$ ; the bounds in the proofs of Lemmas 26, 28 and 29 were tested at 40 000 random rational points of the parameter ranges used there.
- Theorem 4 was confirmed by exhaustive searches over all divisor sets of 21 orders  $p^2q^b$  with  $b \in \{2, 4, 6\}$ , up to  $n = 3^2 \cdot 7^6 = 1\,058\,841$ , with the eigenvalues computed in exact arithmetic from Ramanujan sums, independently of Proposition 6.

- The first proof of Theorem 25 that we found treated Lemmas 28 and 29 by a finer case analysis of the parameters, which reduced them to 137 corners of parameter boxes and 325 polynomial inequalities with nonnegative coefficients, certified by computer. A referee agent re-derived and re-certified these inequalities independently, and all 325 are also proved in Lean. They are not needed for the proof in Section 11.

**Certificates for Proposition 15.** The certificate is implemented in Python, in exact integer arithmetic, in two versions that share the routines building the polynomial matrices. A vectorised version uses 64-bit integers, with an asserted bound that excludes overflow; it first applies the cheaper sufficient test in which the absolute value of each entry of undetermined sign is bounded by the polynomial with the absolute values of its coefficients, and branches over signs only when this test fails. A pure-Python version branches over the signs of the undetermined entries for every sign matrix. The vectorised version certified all eleven exponent pairs; the pure-Python version gave the same result for  $(2, 2)$ ,  $(2, 4)$ ,  $(4, 2)$  and  $(3, 3)$ . An independent re-implementation in SymPy, which builds  $T_k(x)$  from the Ramanujan-sum formula and recomputes the target as  $G(-s_a s_b^T)$ , checking it against the closed form, confirmed  $(2, 2)$ ,  $(2, 4)$ ,  $(4, 2)$  and  $(3, 3)$ . The seven larger exponent pairs were run only with the vectorised version. The polynomial matrices were compared with the exact matrices  $T_k(x)$ ,  $k \leq 8$ , at sample points. As negative controls, the certificate is rejected when the target is lowered by 1, and, for  $(a, b) = (1, 1)$  and  $(2, 2)$ , when the region is enlarged to  $p, q \geq 3$ : at  $p = q = 3$  further sign matrices attain the target. Proposition 15 is not formalised.

**Formal verification.** Theorems 12 and 25, and Theorems 2, 3 and 4 with  $d_k(x)$  taken as  $\|T_k(x)s_k\|_1$  and  $\delta_k(x)$  as  $(T_k(x)s_k)_k$ , are verified in Lean 4 (version 4.33.1, Mathlib commit 0df444a3). The development starts from the definition of the graph and of its energy:

- the adjacency matrix of  $\text{ICG}_n(\mathcal{D})$  has entry 1 exactly when  $\gcd((j - i) \bmod n, n) \in \mathcal{D}$ ;
- the energy is the sum of the absolute values of its eigenvalues, as given by Mathlib's spectral theorem.

It proves:

- Lemmas 7, 8, 10 and 11, and Theorem 12, for real  $p \geq 5$ ,  $q \geq 3$  and all  $a, b \geq 1$ , together with the exchanged form of Remark 13;
- Proposition 6 for all distinct primes  $p, q$  and all exponents, from the eigenvalues of circulant matrices;
- the conclusion of Theorem 2 and the value of the maximal energy, with  $d_k(x)$  defined as  $\|T_k(x)s_k\|_1$ ;
- the conclusion of Theorem 1 for all pairs of distinct odd primes, which includes Theorem 2 and the case  $q \geq 5$ ;
- Theorem 3, as the theorem `ICGEqualParity.corollaryA`. For distinct primes  $p, q \geq 3$  and odd  $m$ , with  $n = pq^m$ ,  $\mathcal{D}^- = \{p^i q^j : i + j \text{ odd}\}$  and  $\mathcal{D}^+ = \{p^i q^j : i + j \text{ even}\}$ , it states that  $\mathcal{D}^-$  and  $\mathcal{D}^+ \setminus \{n\}$  are sets of proper divisors of  $n$ ; that  $2\mathcal{E}(\text{ICG}_n(\mathcal{D}^-)) = n + (3p - 4)\|T_m(q)s_m\|_1 - 2(p - 2)(T_m(q)s_m)_m$ ; that  $\mathcal{E}(\text{ICG}_n(\mathcal{D}^+ \setminus \{n\})) = \mathcal{E}(\text{ICG}_n(\mathcal{D}^-))$ ; and that every set  $\mathcal{D}$  of proper divisors of  $n$  satisfies  $\mathcal{E}(\text{ICG}_n(\mathcal{D})) \leq \mathcal{E}(\text{ICG}_n(\mathcal{D}^-))$ , with equality only if  $\mathcal{D} = \mathcal{D}^-$  or  $\mathcal{D} = \mathcal{D}^+ \setminus \{n\}$ . The case  $n = p^m q$  is the same statement with the primes exchanged. The inequality  $\Phi \leq \Theta$  of Section 9, strict except for the two extremal sign matrices, is proved for all real  $x, R$  with  $x \geq 5$ ,  $R \geq 2$  or  $x = 3$ ,  $R \geq 4$ .

- Theorem 4, as the theorems `theoremB` and `theoremB_swap`. These names, and those in the next item, are in the namespace `ICGEEqualParityB`. For distinct primes  $p, q \geq 3$ , even  $b \geq 2$  and  $n = p^2 q^b$ , `theoremB` states that every set  $\mathcal{D}$  of proper divisors of  $n$  satisfies

$$\mathcal{E}(\text{ICG}_n(\mathcal{D})) \leq \frac{1}{2} [n + (5p^2 - 8p + 4) \|T_b(q)s_b\|_1] - (p^2 - 2p + 2) (T_b(q)s_b)_b,$$

with equality if and only if  $\mathcal{D} = \mathcal{D}^-$  or  $\mathcal{D} = \mathcal{D}^+ \setminus \{n\}$ , where  $\mathcal{D}^- = \{p^i q^j : i + j \text{ odd}\}$  and  $\mathcal{D}^+ = \{p^i q^j : i + j \text{ even}\}$ . The theorem `theoremB_swap` is the same statement for  $n = p^b q^2$ , with the roles of  $p$  and  $q$  in the bound exchanged, and `corollaryB` restates `theoremB` in the format of the theorem for Theorem 3.

- Theorem 25, with its equality cases, as `thmB'`, for real  $p \geq 5$ ,  $q \geq 3$  or  $p = 3$ ,  $q \geq 5$ . There  $G(Y)$  is written as  $\sum_{i=0}^2 \|T_b(q)r_i\|_1 - 2((T_b(q)r_2)_b)^-$ , where  $r_0, r_1, r_2$  are the rows of  $T_2(p)Y$ ; this is the middle expression in (2). The Lean proof follows Sections 10 and 11 with  $a = 2$ . Its main steps are `prop1` (Lemma 21) and `eq_all` (the converse part of Lemma 24) in the module `ICGEEqualParityBMain`; `lemS` (Lemmas 22 and 26) in the module `ICGEEqualParityBCol`; `fd1` (Lemma 28) and `fd2a`, `fd2dom` and `two_col` (Lemma 29) in the module `ICGEEqualParityBFD`; and `G_Yp` (the rest of Lemma 24) in the module `ICGEEqualParityBGraph`, which also contains `thmB'`, `theoremB`, `theoremB_swap` and `corollaryB`. In addition, the 325 polynomial inequalities of the first proof mentioned above are proved in the modules `ICGEEqualParityBCert1` to `ICGEEqualParityBCert6`.

The closed forms of  $d_k(x)$  and  $\delta_k(x)$  in Remark 9 are proved above but not formalised. In the Lean statements,  $d_m(q)$  and  $\delta_m(q)$  appear as `L1 q m sgnv` and `Tv q m sgnv m`, that is, as  $\|T_m(q)s_m\|_1$  and  $(T_m(q)s_m)_m$ ; in the theorems for  $n = p^2 q^b$ ,  $d_b(q)$  and  $\delta_b(q)$  appear in the same way, as `L1 q b sgnv` and `Tv q b sgnv b`. Theorem 5 for  $3 \leq \min\{a, b\} \leq 8$  and Proposition 15 are not formalised. The whole development, including the files for Theorems 3 and 4, was rebuilt from a clean directory. The recorded axiom audits of the main results (the command `#print axioms`) report only `propext`, `Classical.choice` and `Quot.sound`; no native evaluation is used, and the modules were also checked with the kernel replay tool `leanchecker`. The code and logs are available at [1].

## Tool and computational resource disclosure

This work was carried out between 25 and 26 September 2026 in a workflow directed by the author, using the AI coding agents Anthropic Claude Code (model Claude Opus 5.5) and OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`). The author chose the research programme and directed the agents.

- A Claude Code agent found the proof of Theorem 12 and of Theorem 2, formulated Conjecture 14, and wrote the verification programs.
- Another Claude Code agent wrote the Lean formalisation.
- The proof was checked line by line by a second Claude Code agent and by Codex, and the agents searched the literature.
- The results on exponents of equal parity in version 1.2, Theorem 3 and Proposition 15, were found by a Claude Code research agent, and Theorem 3 was checked by an independent Claude referee agent.
- The material added in version 1.4, namely Theorems 4, 5 and 25 and Sections 10–12, was found, formalised in Lean (Theorems 4 and 25) and drafted by Claude Code agents (model Claude Opus 5.5). Theorems 4 and 25 were checked by the Lean formalisation, which

does not cover the case  $3 \leq \min\{a, b\} \leq 8$  of Theorem 5. Earlier write-ups of the new proofs were checked by two independent Claude Code referee agents. One of them checked an earlier write-up of the proof of Theorem 4, which used the 325 certified inequalities mentioned in Section 13; the written proof given here follows the Lean formalisation instead. The other referee agent checked the certificates for the exponents 3 and 4 and an earlier write-up of the computer-assisted proof of Theorem 5, and wrote the second certification program.

- Version 1.5 extends Theorem 5 to  $\min\{a, b\} \leq 8$ . The certification runs for the exponents 5 to 8 were made, and the new text was drafted, by Claude Code agents. A third Claude Code referee agent checked the extension to exponents 5 to 8 and wrote a third certification program.

Claude Code drafted this text. The programs and logs are available at [1]. The author is not a professional mathematician. The author has read version 1.4 of the paper, including Theorems 4 and 5 (the latter then stated for  $\min\{a, b\} \leq 4$ ) and Sections 10–12, and, using a side-by-side table, compared the statements of the Lean theorems with the theorems stated there; the Lean theorems and the statements they formalise are unchanged in version 1.5. The author has also read the changes of version 1.5. The correctness of Theorems 2, 3, 4, 12 and 25 rests on the written proofs and on the Lean formalisation described in Section 13; Proposition 15 and the case  $3 \leq \min\{a, b\} \leq 8$  of Theorem 5 rest on the computer calculations described in Sections 12 and 13; the other independent checks are additional cross-checks. In addition, the arguments for Theorems 2 and 12 were checked by AI agents of two different model families, the argument for Theorem 3 was checked by an independent Claude agent, the earlier write-ups of the proofs of Theorems 4 and 5 were checked by two further independent Claude agents and the extension of Theorem 5 to  $\min\{a, b\} \leq 8$  by a third, and the formal statements of version 1.2 were reviewed by an agent that did not write them. AI systems are not authors of this paper; the author takes full responsibility for its content.

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